

Macroeconomic Forecasting with Text-Based Data

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Abstract

This chapter investigates the potential of using textual data, such as newspaper articles and social media posts, in forecasting macroeconomic variables. It reviews current methods in text mining and textual analysis for transforming unstructured text into numerical indicators useful in macroeconomic nowcasting and forecasting. This review encompasses recent literature on macroeconomic forecasting using diverse text types and highlights the latest applications of large language models in this field. Two empirical applications are presented: the first utilizes sentiment indicators from Reuters news articles in English and two Italian newspapers, *Corriere della Sera* and *La Stampa*, to predict inflation, industrial production, and unemployment in Italy; the second explores predicting inflation expectations using social media posts in Italy updating Angelico et al. (2022).

JEL: C11, C32, C43, C52, C55, E52, E58

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1 Introduction

Macroeconomic forecasting is essential for policymakers, investors, and businesses to make informed decisions. Predicting key economic indicators like GDP growth, inflation, and unemployment rates is challenging due to economic complexity and uncertainties. Traditional econometric models rely on structured numerical data and lagged official indicators for forecasting, which may not capture real-time changes.

To overcome this limitation, researchers have increasingly explored alternative data sources, often referred to as non-traditional or big data. These sources, including textual data from social media, news articles, and official documents, provide real-time and diverse insights into economic sentiment, trends, and market expectations. Leveraging text analysis techniques can complement traditional forecasting models and improve accuracy.

This chapter provides a comprehensive review of text mining methods in macroeconomic forecasting, including sentiment analysis, topic modeling, and the more recent large language models. We explore how these techniques can improve forecasting models, and examine the integration of machine learning, such as deep learning and natural language processing (NLP), to extract meaningful information from textual data.

Our goal is to assess the usefulness of textual data in enhancing established econometric models' forecasting performance. This research contributes to the growing literature on unstructured textual data in economic forecasting and offers insights into the benefits and challenges of incorporating textual variables.

The chapter is structured as follows: Section 2 introduces text mining methodologies, covers data collection, pre-processing, and model specifications. Section 3 reviews econometric models and explores research involving various textual data sources for macroeconomic forecasting, including official documents, news articles, and social media. Section 4 presents empirical results from forecasting exercises using text mining techniques on Reuters and Italian newspaper articles to predict inflation, industrial production, and unemployment in Italy. Additionally, it includes an analysis of Twitter data for inflation expectations in Italy. Finally, Section 5 summarizes the findings and suggests potential avenues for future research.

2 Text as data: overview of Text Mining methods

In this section, we outline key text algorithms and methods for converting unstructured text into numerical formats, integrating them into downstream econometric and forecasting models to predict

macroeconomic variables.

As noted by Ash and Hansen (2023), text analysis in economics has historical roots, evident in early works like Friedman and Schwartz (1963), which constructed policy surprises using manually labeled historical documents. However, the overwhelming volume of text now necessitates machine learning (ML) and deep learning algorithms to interpret text effectively.¹

Ash and Hansen (2023) provide an overview of methods for algorithmic text analysis in economics, focusing on three text representation approaches: (i) high-dimensional count vectors for documents, which represent the frequency of words in a document; (ii) word vectors, which capture semantic meaning of words based on their context; and (iii) embedding vectors for word sequences, which represent sequences of words or sentences in a continuous vector space. They define four key empirical tasks in text-as-data-based economics research: similarity measurement, concept detection, concept relation, and metadata association. The authors explore various research approaches to these tasks, highlighting their respective merits and drawbacks.

2.1 Basic definitions for text analysis

Every text analysis algorithm begins with a machine-readable collection of D documents, known as a ‘corpus’. A document can encompass various texts, from newspaper articles and official filings to social media posts, blogs, or emails.

Defining a document’s scope is crucial. For macroeconomic forecasting using newspapers, a document might be a single article, a paragraph, a title or an aggregation at a desired frequency (daily, weekly, monthly, quarterly). In social media analysis, a tweet (or now, a ‘post’) constitutes a document. This choice impacts analysis granularity, varying from sentence-level to entire documents, influencing algorithm performance.²

Before applying text algorithms, raw document text needs transformation into ‘tokens,’ (i.e. words or word parts) basic linguistic elements in the NLP literature. Denoting a document’s content as $w_d = (w_{d,1}, w_{d,2}, \dots, w_{d,N_d})$, economic applications’ token encoding and pre-processing sequences can differ.

Text mining amalgamates computational linguistics, NLP, statistics, ML, data mining, and information retrieval. Its goal is extracting implicit, previously unknown, and potentially valuable information from extensive text collections.

The text mining process comprises four key stages:

¹See also Jurafsky and Martin (2009), or Gentzkow et al. (2019).

²This choice can also affect the performance of the text algorithms used because some techniques work better with longer documents, while others perform well with shorter ones.

1. **Text Pre-processing:** Conducts syntactic, linguistic, and semantic analyses to prepare text for further processing.
2. **Feature Generation:** Converts text into features or vectors suitable for ML algorithms.³
3. **Feature Selection:** Aims to represent documents using the most pertinent features, reducing dataset dimensions and enhancing ML algorithm performance.⁴
4. **Data Mining:** The final step involves extracting information from the pre-processed corpus using techniques like clustering, classification, or sentiment analysis.

2.2 The typical pre-processing steps

Prior to text analysis, thorough cleaning and pre-processing of raw text data are crucial. This process reduces corpus size and complexity, aiding computational efficiency in subsequent tasks. It involves eliminating grammatically important but contextually insignificant text parts. We divide this into *Text Cleaning*, that is identifying and removing errors, and *Text Pre-processing*, which consists of organizing words for algorithmic interpretation.

Selecting cleaning and pre-processing steps isn't trivial; it heavily relies on the research question, data context, and the application. Ash and Hansen (2023) and Denny and Spirling (2018) emphasize how different combinations of these steps and text algorithms can significantly impact downstream econometric model estimation and inference integrating text-based indicators.

2.2.1 Text Cleaning

Text cleaning aims to eliminate noise and irrelevant information from the text corpus, involving these common tasks:

- **Removing numbers, special characters, and punctuation:** Ensures symbols like commas and periods don't interfere with analysis.⁵
- **Lowercasing:** Standardizes text to lowercase for consistency.

³Also called text transformation or text vectorization.

⁴Feature selection methods encompass four categories: Filter, Wrapper, Embedded, and Hybrid methods. Filters perform statistical analysis on features to select a discriminative subset. Wrappers choose subsets identified and evaluated by classifiers. Embedded methods integrate feature selection into classification training. Hybrid methods combine filter and wrapper approaches.

⁵However, if analyzing at the sentence level or social media text with emoticons, caution is needed as removal could impact subsequent analysis.

- **Eliminating markup language tags and non-textual elements:** Removes HTML, SGML, or XML tags from web-based data to isolate textual content.
- **Spell correction:** Enhances data quality by addressing misspelled words, standardizing abbreviations, and correcting errors like OCR-based issues.⁶

2.2.2 Text Pre-processing

Text pre-processing involves several tasks:

- **Segmentation:** Divides text into sentences, accounting for exceptions like abbreviations (e.g., ‘Inc.’, or ‘Mr.’) or fractional numbers that contain periods.
- **Tokenization:** Splits text into individual ‘tokens’ or words, allowing analysis of terms within the text.⁸
- **Stop Word Removal:** Eliminates common words like ‘the’ or ‘and’ that don’t add significant meaning, reducing dimensionality and enhancing analysis accuracy.
- **Stemming and Lemmatization:** Techniques normalizing words to their root forms, aiding in word variation reduction (e.g., converting “running” to “run” or “better” to “good”).⁹
- **n -grams:** Captures local context information through phrases up to length n often called ‘*collocations*’ (e.g., bigrams like ‘economic growth’ or trigrams like ‘consumer price index’).
- **Part-of-speech (PoS) Tagging:** Assigns word parts of speech (e.g. noun, verb, etc.), although less common in economic applications.
- **Word Sense Disambiguation:** Identifies exact senses of ambiguous words in context, normalizing synonyms or pronouns (e.g., United States, USA, US).

2.3 Feature Generation and Text Vectorization

Following the text pre-processing, researchers transform raw text into numerical vectors for ML algorithms. Common approaches include: i) Bag-of-Words, and ii) Word Embeddings.

⁶This includes fixing typos, garbled text, and Optical Character Recognition⁷ (OCR)-related errors.

⁸For example, the sentence “Economic growth is important” is tokenized as follows: [‘Economic’, ‘growth’, ‘is’, ‘important’].

⁹Stemming removes suffixes, while lemmatization maps words to their base or dictionary form.

2.3.1 Bag-of-Words

The bag-of-words (BoW) model represents text as a collection of words, disregarding grammar and word order but retaining multiplicity. Each word becomes a feature, and the researcher defines the vocabulary with V unique terms from the corpus. Each document is expressed as a vector, through techniques such as **binary term occurrence** or **one-hot encoding**, where Boolean values are used to describe whether a term is present or not in each document. Let $x_{d,v} = \mathbb{1}(w_{d,n} = v)$ be equal to one if the n -th word of document d corresponds to term $v \in V$, for $n = 1, 2, \dots, N_d$ (the number of words in d). Then $x_d = (x_{d,1}, \dots, x_{d,V})$ is the one-hot encoding vector representing d . X is the document-term-matrix (DTM) formed by stacking x_d across rows. X has D rows (number of documents in the corpus) and V columns (number of unique terms in the vocabulary V).

In economics, the number of documents can be significant, such as handling millions of newspaper articles. Even in smaller collections, the number of unique terms can reach tens of thousands. Consequently, these methods result in high-dimensional and sparse DTM.¹⁰

The one-hot encoding technique may not be optimal as word frequency often holds significance in many applications. Hence, alternatives such as **term occurrence**, counting the occurrences of a term in d ($x_{d,v}^{occ} = \sum_{n=1}^{N_d} \mathbb{1}(w_{d,n} = v)$), or **term frequency**, the frequency of the term in d ($x_{d,v}^{tf} = \frac{\sum_{n=1}^{N_d} \mathbb{1}(w_{d,n} = v)}{N_d}$), are commonly employed. However, even with these methods, the resulting DTM can remain high-dimensional and sparse.

Another approach to represent raw text as vectors involves the use of the **term frequency-inverse document frequency** (tf-idf) weighting method.¹¹ This technique computes a word's weight in a particular document, while considering the word's distribution across the entire corpus. The tf-idf weighting is particularly useful in tasks like sentiment analysis. Mathematically, for a document d , the tf-idf is expressed as:¹²

$$x_{d,v}^{tf-idf} = x_{d,v}^{tf} \times idf_v = \frac{\sum_{n=1}^{N_d} \mathbb{1}(w_{d,n} = v)}{N_d} \times \log \left(\frac{D}{\sum_{d=1}^D \mathbb{1}(x_{d,v}^{tf} > 0)} \right) \quad (1)$$

¹⁰According to the historical Oxford English Dictionary (<https://premium-oxforddictionaries-com.libproxy.ucl.ac.uk/words/what-can-corpus-tell-us-about-language>), a comprehensive record of English words includes over 1 million lemmas from past and present usage. To cover 90% of the corpus, 7,000 lemmas are required, and approximately 50,000 to reach 95%.

¹¹This is also related to the "Zipf's law" which states that the frequency of a word in a text is inversely proportional to its rank, with a few highly frequent words and many rare words.

¹²There are different versions of tf-idf - see Manning et al., 2008. Some of these are log-normalized, with the addition of a smoothing value of 1 to account for unseen words or documents, thus avoiding log of zeros.

Here, idf_v adjusts the weight of words that are specific to particular documents, emphasizing rare words while deemphasizing common ones that have survived the stopwords removal (domain-specific stopwords).

The BoW model is a straightforward approach that often yields swift results for basic tasks. When researchers are uncertain about how to initiate the development of text-based leading indicators for macroeconomic forecasting, this simple model serves as a reliable starting point, functioning as an effective baseline.

2.3.2 Word Embeddings

The BoW approach represents documents as global counts over vocabulary terms, regardless of their context, resulting in a highly sparse DTM even with smaller corpora. However, the meaning of each word relies not only on its neighboring words but also on the overall document context. To address these limitations, *word embeddings* represent each word as a dense vector in a low-dimensional space.

Word embeddings focus on local co-occurrences, capturing word context, semantic and syntactic similarity, and word relationships. Similar words are positioned closer in the vector space. For instance, “car” and “automobile” would be nearby vectors, while “car” and “banana” would be distant.

Developed at Google by Mikolov et al. (2013), Word2Vec is a prominent technique for learning word embeddings. It trains words against neighboring words in the corpus using two methods: Continuous Bag-of-Words (CBOW) and Skip-gram.

In CBOW, the objective is to predict a target word w_t based on context words $w_{t-k}, \dots, w_{t+k}$.¹³ Each word w_i is represented as a vector $v_i \in R_E$, where E is the embedding’s dimension (usually 100-500). The context words are averaged to obtain the input context vector $\mathbf{v}_c = \frac{1}{2k} \sum_{i=-k}^k \mathbf{v}_i$. The probability $P(w_t|\mathbf{v}_c)$ of the target word w_t given the context vector $\mathbf{v}_c$ is modeled using the softmax function:

$$P(w_t|\mathbf{v}_c) = \frac{\exp(\mathbf{v}'_t \mathbf{v}_c)}{\sum_{i=1}^W \exp(\mathbf{v}'_i \mathbf{v}_c)} \quad (2)$$

Here, W represents the vocabulary size, and $\mathbf{v}'_i$ is the output vector for word i .

The Skip-gram formulation reverses this to predict context words given an input word w_t :

$$P(w_{t-k}, \dots, w_{t+k}|w_t) = \prod_{-k \leq j \leq k, j \neq 0} P(w_{t+j}|\mathbf{v}_{w_t}) \quad (3)$$

¹³ k is the context window, typically ranging between five and ten words for Word2Vec models.

GloVe and FastText are two other notable word embedding techniques. Developed at Stanford by Pennington et al. (2014), GloVe generates embeddings by factorizing a matrix of word co-occurrences. Introduced at Facebook by Grave et al. (2018) and Joulin et al. (2018), FastText offers pre-trained embeddings for over 150 languages and differentiates itself by incorporating subword information like n-grams, enhancing its ability to handle morphological nuances and out-of-vocabulary words.

2.3.3 Contextual Embeddings

Traditional word embeddings like Word2Vec derive word representations based on local context but lack global contextual sensitivity. These embeddings remain fixed for each word regardless of its context within a sentence or document, offering limited adaptability to different meanings.

In contrast, contextual embeddings, exemplified by ELMo (Embeddings from Language Models) by Peters et al. (2018) and BERT (Bidirectional Encoder Representations from Transformers) by Devlin et al. (2019), revolutionize word representation by considering dynamic contextual variations. ELMo utilizes a bidirectional Long Short-Term Memory (LSTM) network, capturing nuanced contextual details by analyzing text from both directions, generating multiple layers of embeddings that range from detailed syntactic information in lower layers to abstract semantic meanings in higher layers.

BERT further enhances contextual embeddings through a two-stage training process. Initially pre-trained on vast text corpora using tasks like masked language modeling and next-sentence prediction, BERT acquires profound contextual understanding. This pre-training equips BERT with adaptable embeddings that encode rich semantic and syntactic information, offering versatility for diverse downstream applications.

2.4 Data mining: Text analysis tasks

Once a text corpus is cleaned and pre-processed, various text analysis techniques can be applied to extract meaningful information. These include:

- **Sentiment Analysis:** This involves determining the tone or sentiment of the text, such as positive, negative, or neutral. It's useful for understanding opinions or attitudes expressed in the corpus.
- **Topic Modeling:** This technique identifies themes or topics within the text. It's beneficial for uncovering underlying patterns in large text datasets.

- **Other Tasks:** There are additional tasks relevant to economic research, such as Named Entity Recognition (NER) (identifying and classifying key information like names, organizations, locations),¹⁴ and Syntax Parsing (analyzing the grammatical structure of sentences).

Each of these tasks offers unique insights and contributes to a comprehensive understanding of the textual data, supporting various applications in economic research. However, we will only discuss sentiment analysis and topic modeling which are more frequent in macroeconomic forecasting.

2.4.1 Sentiment Analysis

Sentiment analysis evaluates the emotional tone of text, crucial for understanding public opinion on economic topics and analyzing sentiment in financial news. This process involves diverse methodologies, from simple rule-based or lexicon-based approaches to advanced ML and deep learning algorithms (Text Classification).

A common approach in lexicon-based sentiment analysis is to calculate sentiment and polarity scores using a predefined lists of words, annotated with their sentiment polarities (positive, negative, or neutral). For instance, if P , N , and T represent the count of positive, negative, and total words in a document, respectively, the sentiment score of a document i can be calculated as:

$$S_i^1 = \frac{P - N}{T}$$

and the polarity score as:

$$S_i^2 = \frac{P - N}{P + N}$$

An example of these methods is VADER (Valence Aware Dictionary and sEntiment Reasoner) proposed by Hutto and Gilbert (2014) which assigns a polarity score to each word, also considering semantic rules such as negation and modifiers (valence shifters) to adjust sentiment and capture the emotions expressed in the text. Similarly, the AFINN dictionary by Nielsen (2011) assigns polarity scores to words. Both VADER and AFINN are trained on social media posts and, therefore, they might not capture nuances of sentiment specific to economic or finance contexts,¹⁵ which require tailored approaches. Loughran and McDonald (2011) (LM) and Correa et al. (2020) address this by developing, respectively, the LM and financial stability (FS) dictionary that are more suited for

¹⁴NER typically employs ML and specialized NLP libraries like spaCy.

¹⁵Actually, while being neutral in the economic/financial domain, in a general context, words like ‘profit’ or ‘growth’ may be considered positive, and ‘risk’ or ‘loss’ may be labeled as negative.

financial contexts.¹⁶

Another family of sentiment classifiers comes from applying ML techniques to classify texts. These techniques include Naive Bayes, Support Vector Machines (SVM), Logistic Regression, Recurrent Neural Networks (RNNs), and transformers like BERT. Nevertheless, this approach requires: (i) fitting a model to the data, and (ii) having human-labeled text.

In economic research, sentiment analysis has been effectively applied for forecasting. For instance, Shapiro et al. (2020) developed a sentiment measure from economic and financial newspaper stories to analyze its relationship with economic indicators. Their approach combines existing lexicons and accounts for negation, yielding a model with high predictive accuracy. Aprigliano et al. (2023) demonstrate the application of sentiment analysis in macroeconomic forecasting. They compute sentiment indicators from Italian newspaper articles using a novel Italian economic dictionary, showcasing the importance of context-specific sentiment analysis in economic research. Consoli et al. (2022) propose FiGAS, a Fine-Grained Aspect-based Sentiment analysis method, which is customized for the economic/financial domain and can be applied to specific topics/words of interest in a sentence, such as inflation or GDP growth. The authors applied this method to ECB bulletins and FED Beige Books, showing that FiGAS can predict recessionary periods better.

These examples highlight the significant role of sentiment analysis in economic research and forecasting, utilizing both lexicon-based and ML approaches to derive meaningful insights from textual data.

2.4.2 Topic Modeling

Topic modeling is a powerful tool in text analysis, especially useful in economics for uncovering underlying themes or narratives in large text corpora, such as newspapers or public discourse. This unsupervised learning approach does not require labeled data, relying instead on the co-occurrence of words within the corpus to identify relevant topics.

A prominent algorithm in topic modeling is Latent Dirichlet Allocation (LDA) by Blei et al. (2003). LDA posits that each document is a blend of various topics, and each topic is characterized by a distribution of words. The algorithm iteratively reassigns words to topics, considering their probability distributions, to uncover the latent topics that likely generated the collection of documents.

While LDA operates primarily on the text corpus, economic research often has accompanying

¹⁶Recently, Barbaglia et al. (2022) suggest an Economic Lexicon (EL) which offers a comprehensive coverage of economic terms and includes human-annotated sentiment scores, facilitating nuanced sentiment analysis. Compared to existing lexicons, the EL demonstrates superior performance.

metadata (like publication dates, or macroeconomic conditions). Advanced models, such as the Structural Topic Model (STM) by Roberts et al. (2013), incorporate this metadata to influence topic prevalence and content. This allows for nuanced analysis of how topics are discussed under varying conditions.

Non-Negative Matrix Factorization (NMF) is another approach that factorizes DTM into two matrices representing documents and terms, respectively. This technique is particularly suited to word frequency data due to its non-negative constraint.

Interpreting identified topics typically involves examining words and documents with high probabilities in each topic. Researchers often label these topics based on prevalent words, elucidating key themes in the corpus.

Bybee et al. (2024) demonstrate the application of LDA in summarizing US business news into thematic topics, correlating news attention with economic activities and stock market forecasts. Conversely, Kalamara et al. (2022) caution against using LDA or similar models for forecasting to avoid look-ahead bias and computational challenges. They explain that using entire samples for topic estimation can bias forecasts due to future-informed term counts. To mitigate this, Bybee et al. (2024) recommend the Online LDA (oLDA) estimator by Hoffman et al. (2010), which constructs topic models in a backward-looking manner, thus avoiding look-ahead bias.

2.5 Large Language Models

Recently, Large Language Models (LLMs)¹⁷ are revolutionizing NLP. An LLM is a deep learning-based neural network architecture characterized by a substantial number of parameters trained on extensive datasets, which enables it to perform various NLP tasks such as text generation, translation, and sentiment analysis with high levels of accuracy.

Economists are increasingly using LLMs. Korinek (2023) identify 25 use cases across domains, suggesting that LLMs can automate micro-tasks and boost research productivity. However, Buchanan and Shapoval (2023) find limitations in precision and citation accuracy, emphasizing the need for careful fact-checking.

Recent NLP advancements integrate transfer learning with LLMs. This approach enhances NLP capabilities by pre-training models on general tasks and fine-tuning them on specific datasets, addressing scarcity of labeled data and computational costs.

Despite their promise, LLMs have limitations like perpetuating biases, limited context under-

¹⁷Such as BERT, GPT3, GPT3.5, GPT4 (Generative Pre-trained Transformer 3, 3.5, and 4), LLaMA (LLM Meta AI), Vicuna, Palm, Orca, etc.

standing, opacity in decision-making, and computational demands. Responsible development and ongoing research are essential to address these challenges.

3 Macroeconomic forecasting with text as data: models and literature

In this section, we will examine the conventional econometric models utilized for nowcasting/forecasting macroeconomic variables like GDP growth, inflation, and unemployment. Additionally, we will explore literature employing various textual sources: Google Trends, official documents, newspapers, or social media.

3.1 Forecasting models

Macro forecasting models are typically enhanced with leading indicators derived from non-traditional data, such as textual data, to augment their out-of-sample performance. Being more timely, these indicators incorporate information that bolsters the predictive capacity of several models.

There are various macroeconomic forecasting models for a target variable y_t at time t , each possessing its own set of strengths and weaknesses. For each model, we will present the direct h -step-ahead forecasts $\hat{y}_{t+h}$ conditioned on the information set at time t , denoted as I_t . The most prevalent forecasting models include:

- *Random walk* (RW): The simplest model assumes future values equal the current value plus a random error. Easy to implement and used as a benchmark, yet less accurate for long-term predictions. Its h -step-ahead forecast is given by:

$$y_{t+h} = y_t + e_{t+h}, \quad h \geq 1 \quad (4)$$

where e_{t+h} represents a random error term.

- *Autoregressive models* (AR) use past variable values to predict future ones. They can outperform RW in short-term forecasting and serve as benchmarks. The h -step-ahead forecast from an $AR(p)$ is given by:

$$y_{t+h} = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + e_{t+h} \quad (5)$$

- *Autoregressive Moving Average (ARMA) models* combine AR and Moving Average (MA) components, utilizing a linear combination of error terms at various lags. The h -step-ahead forecast of an $ARMA(p, q)$ is given by:

$$y_{t+h} = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{i=1}^q \theta_i e_{t-i} + e_{t+h} \quad (6)$$

- *Static factor models (SFM)* or diffusion indexes (DI) use a few common factors from a set of macroeconomic variables to explain their variation. Stock and Watson (2002) demonstrate that its h -step-ahead forecast is expressed as:

$$y_{t+h} = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{k=1}^K \beta_k F_{k,t} + e_{t+h} \quad (7)$$

$$X_t = \Lambda F_t + \varepsilon_t \quad (8)$$

where F_t are $K \ll N$ common factors estimated via principal components from N macro variables X_t , and Λ represents factor loadings.

- *Dynamic factor models (DFM)* resemble SFM but factors evolve over time. The h -step-ahead forecast of a DFM, akin to (7)-(8), incorporates an additional aspect: the latent factors follow a VAR expression:

$$F_t = AF_{t-1} + \eta_t \quad (9)$$

where $\eta_t \perp \varepsilon_t$.

- *Mixed Data-Sampling (MIDAS) models* by Ghysels et al. (2007) utilize mixed-frequency data, (daily, weekly, etc.). For instance, a MIDAS equation predicts quarterly GDP growth y_{t+h} using lags y_{t-j} and a high-frequency indicator x_{t+w}^m (e.g., monthly industrial production or Purchasing Manager Index - PMI). Conditioned on the current quarter information, the equation is:

$$y_{T_y+h|T_x} = \alpha + \phi y_{T_y} + \beta_1 B(L^{1/3}; \theta) x_{T_x}^m + e_{t+h}, \quad \text{with} \quad (10)$$

$$B(L^{1/3}; \theta) = \sum_{k=1}^K b(k; \theta) L^{k/3} \quad (11)$$

where $B(L^{1/3}; \theta)$ is a lag polynomial, L is the lag operator, $w = T_x - T_y$ indicates the high-frequency indicator's lead, and the monthly lag operator is defined as $x_{t-1/3}^m = L^{1/3} x_t^m$.

- *Machine Learning (ML) models*: Recent interest focuses on utilizing various ML techniques like Lasso, Ridge, Random Forest, Neural Networks, or ensemble methods for macroeconomic forecasting.¹⁸ Due to their non-parametric and non-linear nature, these models often outperform traditional ones. In its basic form, the h -step-ahead forecast is expressed as:

$$y_{t+h} = f(y_{t-1}, \dots, y_{t-p}) \quad (12)$$

where f varies depending on the ML algorithm used.

These forecasting models can be enhanced by incorporating leading indicators Z_t to improve nowcasting/forecasting accuracy. Surveys from households and businesses, known as soft indicators, are often used to assess if they offer better forecasts than macroeconomic variables usually available with some delay. For instance, an AR-X model using PMI to predict GDP growth is represented as:

$$y_{t+h} = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + \gamma Z_t + e_{t+h} \quad (13)$$

With the emergence of non-traditional data, Z_t is often characterized by high-frequency updates, on a daily or even intra-daily basis, requiring aggregation or averaging. Text-based leading indicators, denoted as S_t , can contribute to the prediction of an SFM:

$$y_{t+h} = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{k=1}^K \beta_k F_{k,t} + \gamma S_t + e_{t+h} \quad (14)$$

Forecasts from these models are compared against one or more benchmarks to evaluate the efficacy of the new text-based indicators. However, to compute these indicators and convert textual

¹⁸See for example Medeiros et al. (2021), Masini et al. (2023), or Babii et al. (2022).

data into time series, researchers face several crucial choices that impact their forecasts:

1. The choice of *corpus*: Whether to use the entire dataset (e.g., all articles from a newspaper) or a selection based on predefined keywords or lexicons.
2. Selection of *language*: Choosing between English or another language (e.g., Italian, French, or Chinese), leading to decisions regarding translations (which introduces some form of bias) or creating specific dictionaries.
3. *Aggregation level*: Deciding the granularity of analysis, ranging from title, sentence, paragraph to full document for information extraction from newspaper articles.
4. *Pre-processing* steps: Determining the necessary pre-processing techniques.
5. *Text representation model selection*: Choosing among various models, including dimensionality reduction, modern NLP algorithms (e.g., word embeddings, BERT), or LLMs.

3.2 Forecasting using Google Trends: Textual data without Text Mining

Google Trends (GT) has emerged as a pivotal tool in macroeconomic forecasting, providing a unique approach to text-based indicators without the complexity of text mining. GT¹⁹ offers the SVI for specific keywords, which is normalized and standardized for comparability. This data allows researchers to gauge public interest in various topics worldwide or people intentions, providing valuable insights for economic forecasting.

Pioneering studies in this domain have effectively used GT data for forecasting unemployment rates. For example, Askitas and Zimmermann (2009) and D’Amuri (2009) utilized GT (at that time called Google Insights) to predict German and Italian unemployment rates, respectively. Further studies by D’Amuri and Marcucci (2009), and D’Amuri and Marcucci (2017) demonstrated that GT SVI for job-related keywords enhances the predictive power of time series models for the US unemployment rate, surpassing traditional indicators like initial claims.

The utility of GT in forecasting other macroeconomic variables has been corroborated by various studies. Vosen and Schmidt (2011) and Woo and Owen (2019) found that Google search data outperforms traditional survey-based indicators in predicting US private consumption. Additionally,

¹⁹The Search Volume Index (SVI) for specific keywords or categories and sub-categories (groups of related keywords) can be downloaded directly from <https://trends.google.com/trends/> or with APIs. The tool provides the relative frequency with which a particular keyword is searched for in several languages from various countries at frequencies that range from one minute to one month.

Götz and Knetsch (2019) and Ferrara and Simoni (2023) showed that Google search data enhances the nowcasting accuracy of GDP growth in Germany, the Euro Area, and the US, particularly during recessions.²⁰ Niesert et al. (2020) explore the use of Google search data to predict unemployment, consumer price index (CPI), and consumer confidence for several countries, finding that such data is valuable only in predicting unemployment. In fact, the predictive value of GT data may vary depending on the specific macroeconomic variable and its relationship with user search behavior. In the same spirit, Bantis et al. (2023) explored the effectiveness of GT data in forecasting GDP growth in the US and Brazil, finding that factor models incorporating economic indicators and Google “categories” outperform in both countries. However, using “sub-categories” was found to be not advantageous.

Recent advancements include Borup and Schütte (2022), who demonstrated that GT is a valuable tool for forecasting employment growth in the US at various horizons and at a more granular level. Woloszko (2023) further extends the application of GT categories/sub-categories by developing a weekly economic activity tracker for 46 OECD countries, effectively capturing economic dynamics during critical periods like the COVID-19 lockdowns. Stolbov and Shchepeleva (2023) have introduced sentiment-based indicators of real estate market stress using GT data, illustrating its predictive power. Finally, Kohns and Bhattacharjee (2023) explored the benefits of GT data for nowcasting real US GDP growth, introducing advanced methodologies that improve the accuracy of forecasting models.

In summary, GT data has proven to be an invaluable asset in macroeconomic forecasting. Its ability to provide real-time insights into public interest and intentions, coupled with its global reach and applicability, underscores its potential as a powerful alternative to traditional forecasting methods.

3.3 Forecasting using Official Documents

In this section, we delve into the growing interest in utilizing official documents, such as central bank communications, corporate earnings calls, and other public texts, to enhance economic forecasting.²¹ Several studies have explored the impact of text analysis on economic predictions using these resources.

Central bank communications, particularly the Federal Open Market Committee (FOMC) minutes, have been a focal point. Ericsson (2016) and Stekler and Symington (2016) introduce quanti-

²⁰Unlike GT, Google search data represent volume fluctuations of Google queries relative to the initial value.

²¹While central bank communications are often meticulously curated to convey precise messages, corporate earnings calls transcripts may be less curated, yet potentially more candid in relation to the target variable of interest.

tative sentiment indexes from FOMC minutes, offering early insight into Fed Greenbook forecasts. Hansen and McMahon (2016) use computational linguistics to assess the impact of FOMC statements on financial markets and the macroeconomy, emphasizing the importance of forward guidance. Ferreira (2021) and Lima et al. (2021) leverage FOMC statements to enhance macroeconomic forecasts. Baranowski et al. (2021) investigate the influence of central bank communication, interest rates, and projections on private sector expectations in Poland.

The Bank of England’s Quarterly Inflation Reports with qualitative narratives are examined by Clements and Reade (2020) and Jones et al. (2020), emphasizing the value of narratives in improving forecasting.

The Federal Reserve’s Beige Book proves useful for GDP and employment forecasts, as shown by Armesto et al. (2009). Sadique et al. (2013) investigate the tone of language in the Beige Book for business cycle forecasting, and Balke et al. (2017) explore innovative ways to integrate textual data into economic forecasts.

Earnings call transcripts can offer valuable information. Gosselin and Taskin (2023) and Windsor and Zang (2023) use sentiment from earnings calls to forecast GDP growth, inflation, and other economic indicators, for the US, Canada and Australia, respectively.

Textual data from surveys and institutional reports also play a role. Nakajima et al. (2021) develop a quantitative indicator of firms’ inflation expectations in Japan. Cajner et al. (2024) analyze U.S. manufacturing firms’ survey responses to improve industrial production forecasting. Müller (2022) converts German forecasters’ narratives into sentiment indices for improved business cycle forecasts.

In conclusion, text analysis of official documents has significantly advanced economic forecasting. These studies collectively highlight the untapped potential within textual data, offering valuable tools for policymakers and economists.

3.4 Forecasting using Newspapers

In macroeconomic forecasting, leveraging newspaper articles has become pivotal for anticipating economic trends. These articles provide insights that align with the broader public’s perception, which influences decision-making and economic behavior. Researchers have explored various techniques to harness the real-time information within these articles to integrate emotional tones from news into economic forecasts. These approaches demonstrate the potential of newspapers to capture nuanced economic signals, enhancing nowcasting accuracy. The choice of text granularity, language, and algorithm plays a critical role in the effectiveness of this approach.

Studies like Kalamara et al. (2022), Aprigliano et al. (2023), Lunde and Torkar (2020), and Thorsrud (2020) highlight how simple text-based indicators and sentiment analysis from newspapers enhance economic forecasts in UK, Italy, China and Norway, respectively, especially during downturns. These studies demonstrate the effectiveness of combining text counts with ML techniques, such as supervised models and LDA or context-aware dictionary-based methods, to capture public sentiment and economic trends. Similarly, other papers confirm the forecasting ability of text-based indicators derived from newspapers in single countries: for example, Bortoli et al. (2019) for France, Ellingsen et al. (2022) and Barbaglia et al. (2023) for the US, Goshima et al. (2021) for Japan, Rambaccussing and Kwiatkowski (2020) for the UK, Ulbricht et al. (2017) for Germany, or Zheng et al. (2024) for China.

A global perspective is evident in works like Elshendy and Fronzetti Colladon (2017), Ashwin et al. (2021) and Barbaglia et al. (2024), which analyze economic news across various countries. These studies underscore the importance of considering different languages (even if often translated into English) and regional contexts in sentiment analysis. They reveal how customized sentiment dictionaries and fine-grained text analysis, tailored to economics, can significantly improve GDP forecasts.

The role of newspapers in shaping public perception and expectations is explored in studies like Larsen et al. (2021) and Tilly et al. (2021). These works illustrate how news topics and sentiments can influence inflation expectations and information rigidity among households, offering insights into the relationship between media coverage and economic perception. Tilly and Livan (2022) and Larsen and Thorsrud (2019) show how the extraction of specific emotions and news topics enhances the predictive power for various economic indicators. Recently, van Dijk and de Winter (2023) expand the scope of newspaper-based forecasting by introducing a tone-adjusted time-varying layered LDA with time-varying hierarchical topics.

The effectiveness of newspapers in providing real-time insights into economic conditions is further supported by Lukauskas et al. (2022), Ferrari Minesso et al. (2023), and Ardia et al. (2019). These studies show how daily sentiment metrics and text-based indicators can serve as reliable predictors for various economic activities, from GDP growth to inflation rates.

Collectively, these studies provide compelling evidence that newspapers, through sentiment analysis, topic modeling, and real-time data extraction, are indispensable tools in macroeconomic forecasting. They highlight the adaptability of these methods across different languages and economic contexts, proving particularly effective during periods of economic uncertainty. This body of work underscores the significant potential of leveraging newspapers for more accurate and timely economic forecasts globally.

3.5 Forecasting using Social Media or Internet Data

Social media and internet data have emerged as potent tools for macroeconomic forecasting, offering real-time insights that traditional surveys struggle to match. This section connects a range of studies that demonstrate the utility of social media data in measuring and forecasting economic variables, highlighting the global scope and diverse applications of this approach.

Social media platforms, particularly Twitter (now called X), have been extensively utilized due to their public APIs. Studies like Indaco (2020) and Antenucci et al. (2014) highlight how Twitter data can be used to supplement official economic indicators like GDP estimates and unemployment insurance claims. The granularity of social media data allows for precise estimations at sub-national levels, making it a valuable resource for both governmental and non-governmental agencies.

The predictive power of social media sentiment is further exemplified in Angelico et al. (2022), who demonstrate the use of Twitter data to measure and forecast inflation expectations in Italy. Using recent advancements in NLP, Born et al. (2023) and Munoz et al. (2023) have developed more nuanced economic indices, showing how transformer models and ChatGPT can extract sophisticated sentiment indicators from Twitter data, closely mirroring actual inflation trends and consumer expectations.

The potential of online platform data, beyond traditional social media, is explored in Glaeser et al. (2022), showing that Yelp data can be a powerful supplement to official datasets for gauging local economic activity in real-time.

In the context of the real estate market, Berka and Mao (2023) demonstrates the significance of social media sentiment from Weibo in predicting house price changes in major Chinese cities.

Expanding the scope to image-based social media data, Indaco (2023) showcases how image tweets can enhance GDP estimates, particularly in developing countries. The study’s findings highlight the versatility and breadth of social media data in economic forecasting.

Furthermore, Grebe et al. (2024) and Gallego et al. (2020) illustrate the application of social media data in understanding the impact of geopolitical events, such as the war in Ukraine, and regional GDP forecasting, respectively. These studies utilize advanced text analysis methods and deep learning to extract meaningful economic indicators from social media content.

Lastly, Ayivodji (2020) and Pokhriyal et al. (2022) highlight the use of big data and ML to develop high-frequency, real-time indicators of consumer inflation expectations and labor market flows. These approaches underscore the evolving landscape of economic forecasting, leveraging the immediacy and geographical precision of social media data.

In summary, the integration of social media and internet data into economic research represents

a significant paradigm shift. These platforms offer real-time, geographically precise data that complement traditional survey methods, enabling more efficient and accurate economic forecasting. The diverse global applications of this data, ranging from GDP estimation to understanding inflation perceptions and labor market dynamics, illustrate its broad relevance and potential in contemporary economic analysis.

3.6 Forecasting using Large language models

There is an emerging literature that analyzes the potential of LLMs to undertake macroeconomic and financial forecasting.

For example, Faria-e-Castro and Leibovici (2023) explore the potential of LLMs for generating inflation forecasts. Using Pathways Language Model (PaLM), they produce retrospective conditional forecasts, comparing them to the SPF. They find the LLM forecasts yield lower Mean Squared Errors (MSEs) overall and for nearly all forecast horizons. LLM forecasts also exhibit slower reversion to the 2% inflation anchor compared to the SPF. The study suggests LLMs may offer inexpensive and accurate alternatives for forecasting with wide applicability. However, the authors note that LLMs operate as black boxes, making it difficult to fully understand the mechanics behind their output.

In finance, Yu et al. (2023) use GPT-4 and Open LLaMA for explainable financial time series forecasting, finding that LLMs outperform both ARMA-GARCH and ML algorithms in predicting the Nasdaq100. LLMs also provide reasoned explanations for their forecasts by extracting insights across time series.

This literature is evolving very fast given the continuous improvements with Generative AI, foundations models and LLMs. We believe that in a few years, an increasing amount of research papers will use LLMs and Generative AI to produce more and more outperforming leading indicators based on different sources of text.

4 Two empirical applications of text-as-data for macro forecasting

4.1 Predicting Italian macro variables Using Newspapers' sentiment

In this section, we explore the empirical results from an extensive forecasting exercise. This study utilized a dataset comprising 220,000 Reuters news articles in English, covering economic developments in Italy from 1987 to 2023. Additionally, we analyzed 135,000 articles in Italian from

two leading newspapers, *Corriere della Sera* and *La Stampa* from 1997 to 2023.²² These articles were processed through several text analysis steps to extract various text-based indicators. These indicators were then integrated to forecast key macroeconomic variables in Italy.

Both data corpora were sourced from Dow Jones Factiva Analytics, where we gathered all Reuters articles with a focus on economic topics²³ pertaining to Italy. To ensure comprehensive coverage, we extracted the entire text, specific paragraphs containing noteworthy economic news about Italy (paragraphs containing “ital ”), and article titles.²⁴

In Figure A.1, we display word cloud visualizations derived from the articles and paragraphs, highlighting how narratives have evolved through two distinct pre-processing stages: (i) removal of stopwords, punctuation, and digits; and (ii) lemmatization. By revealing variations in narratives across the articles and paragraphs, these word clouds help to illustrate potential impacts of different pre-processing steps on the leading indicators in downstream predictions.

Figure A.2 offers insights into sentiment indicators computed through the application of four different lexicons (VADER, FS, LM, and AFINN) across Reuters articles, paragraphs, and titles. Additionally, we include sentiment analysis from economic articles and titles in Italian newspapers (*Corriere della Sera* and *La Stampa*), as in Aprigliano et al. (2023) where sentiment is computed using Bruno et al. (2018)’s Italian dictionary. Notably, sentiment extracted from paragraphs and titles demonstrates a higher level of precision, revealing distinct patterns such as prominent negative sentiment spikes during pivotal events like the COVID-19 pandemic and the sovereign debt crisis.

To assess forecasting accuracy, we employed two modeling approaches: benchmark AR(p) models and augmented AR-X(p) models that incorporated sentiment indicators from Figure A.2. The forecasting exercise was conducted on the common sample from 1997 to 2023, with the out-of-sample spanning from July 2007 to April 2023. Figure A.3 showcases the Cumulative Sum of Squared Error Differences (CSSED) between benchmark and competing models in predicting Italian inflation across various forecast horizons. We employed different English lexicons for sentiment computation and observed variations in performance, with FS and LM dictionaries exhibiting stronger performance, especially after 2022.

Figure A.4 illustrates the CSSED for forecasts of Italian industrial production. Here, VADER and AFINN sentiment outperformed at shorter horizons, while FS and LM delivered superior results

²²We selected economic articles as in Aprigliano et al. (2023).

²³We downloaded articles with the following news subjects: Eurozone Data/Policies (e52), Monetary Policy (e12), Economic Performance/Indicators (e11), Foreign Direct Investment (efdi), Economic News (ecat), Purchasing Manager Index (epmi).

²⁴This data collection process was facilitated using PySpark and SQL to select the articles and Python for text-mining. In particular, we used the libraries: nltk, gensim, spaCy, textstat, wordcloud, sklearn, re, bs4, and request.

at 12-month forecast horizons, particularly during the challenging period of the Great Financial Crisis.

Figure A.5 presents CSSED for Italian unemployment rate predictions. Surprisingly, VADER and AFINN sentiment extracted from Reuters articles and paragraphs consistently outperformed other lexicons, while LM and FS showed improved performance towards the latter part of the out-of-sample period. It’s noteworthy that models utilizing FS sentiment on paragraphs surpassed those utilizing LM.

To summarize, text-based forecasts derived from Reuters excel over longer forecast horizons and across various document types. Conversely, predictions relying on Italian newspapers exhibit more variability, although forecasts based on sentiment from *Corriere della Sera*’s titles appear to outperform. Sentiment indicators computed using FS or LM dictionaries consistently outperform others, occasionally favoring FS. Context-specific sentiment scores from paragraphs outperform full articles. Unemployment forecasts consistently outperform, while results for inflation and industrial production display mixed performance, with text-based forecasts for inflation particularly shining in the latter part of the out-of-sample period.

These empirical findings underscore the critical role of meticulous data source selection, language preferences, and thoughtful algorithm choices when creating text-based leading indicators for economic forecasting, echoing the observations made in Ash and Hansen (2023).

4.2 Predicting Italian Inflation Expectations Using Twitter Data

In this section, we present updated forecasts originally proposed by Angelico et al. (2022), predicting Italian inflation expectations using Twitter data. We extended the dataset from June 2013 to April 2023, encompassing 23 million Italian tweets by approximately 1.6 million users. Notably, the daily volume of tweets concerning inflation and price dynamics surged from 2,500 in 2012 to 12,000 in 2022 due to events like the COVID-19 pandemic and the Ukraine conflict, as shown in Figure A.6. We followed the same methodology as Angelico et al. (2022) to filter out advertisements and create inflation expectation indicators, including LDA topic modeling without reestimating the model. As shown in Figure A.7, these indicators exhibit a strong and significant correlation with monthly survey-based expectations from the Italian National Statistical Office (ISTAT), despite the pandemic and energy crisis in 2020 and 2022. A similar pattern emerges when comparing daily Twitter-based indicators with the one-year inflation swap (Figure A.8).

Our out-of-sample forecasting exercise, extending until April 2023, builds on the original study, with the first in-sample window running from June 2013 to May 2016. Predicting Italian survey-

based inflation expectations from one to six months ahead, we employ various forecasting models and include a GT-based indicator aligned with the keywords used to select the raw tweets (see Table A.1). Contrary to the original sample, only Twitter-based indicators from accounts interested in economics (bios containing ‘Econ’) yield relatively strong forecasting results across all horizons.²⁵ However, it’s worth noting that these results aren’t surprising, as most recent macro forecasting papers either exclude or treat the COVID-19 pandemic differently.

Figure A.10 illustrates the CSSED for forecast horizons from one to six months when using a 30-day backward-looking MA to smooth Twitter-based indicators from users interested in economics. The GT indicator consistently hovers around zero, indicating limited outperformance. The market-based index shows some predictive ability towards the end of 2021 and early 2022, but Twitter-based indicators tend to outperform during this period. Notably, TW2-MA(30), accounting for volume spikes related to inflation releases and specific events, consistently outperforms other indicators, even in late 2022 and early 2023. However, it’s essential to consider that the acquisition of Twitter by Elon Musk in October 2022 led to significant user departures, and substantial decreases in volumes, potentially affecting indicator accuracy.

Figure A.11 shows the CSSED when using Twitter-based indicators from all users. These indicators demonstrate outperformance since 2018, but some lose forecasting ability in late 2022 and 2023. Remarkably, TW2-MA(30) maintains consistent outperformance throughout the entire out-of-sample period.

To summarize, Twitter-based indicators for inflation expectations demonstrate robust predictive performance throughout the out-of-sample period, including during the challenging times of the COVID-19 pandemic and the initial phase of the 2022 energy crisis. However, it’s noteworthy that many indicators appear to lose their predictive ability towards the end of the out-of-sample period, coinciding with the acquisition of Twitter. This could also be due to the fact that the LDA model was not reestimated.

Textual data, such as that derived from social media platforms, offers valuable insights for economic forecasting. It provides a real-time and large-scale source of information, allowing researchers to capture evolving sentiment and public opinion. Additionally, text-based indicators can complement traditional economic data, potentially enhancing forecasting accuracy.

²⁵Figure A.9 shows that indicators obtained from ‘Econ’ or ‘News’ users are highly correlated with those obtained from the whole sample. Nevertheless, indicators from accounts interested in news (bios containing ‘News’) perform slightly worse than ‘Econ’ especially at short horizons.

5 Conclusion

This chapter underscores the substantial potential and inherent challenges of utilizing textual data in macroeconomic forecasting. While sentiment indicators derived from text mining, such as those from newspaper articles and social media posts, have shown promise in predicting crucial economic variables like inflation and unemployment, the approach is not without its complexities.

The predictive utility of text-based metrics can vary significantly, influenced by methodological choices, data sources, linguistic differences, and the specific economic context of different countries and periods. The risks associated with overfitting sparse text data and the replication difficulties of complex models underscore the need for a balanced approach that combines text mining, feature engineering, and econometric modeling.

While textual data offers timely insights and captures emerging trends, it also presents challenges such as potential data biases and sensitivity to factors like changes in user behavior or platform policies.²⁶ Researchers must navigate these complexities to effectively leverage text data in economic analysis and forecasting.

Nevertheless, the empirical studies and literature reviewed in this chapter compellingly demonstrate the value of text data in enhancing forecasting accuracy and providing real-time economic insights. The future of macroeconomic forecasting with text data lies in the responsible development and integration of advanced NLP techniques, continuous research, and expansion of text corpora.²⁷

In closing, text-as-data techniques provide a unique opportunity to integrate alternative data sources into economic forecasting for more accurate, granular, and timely insights. Despite the challenges, further research and development in this field are essential, promising enhanced tools for policymakers, forecasters, and researchers in understanding and predicting economic dynamics.

²⁶Cao et al., 2023 demonstrate that firms strategically alter the linguistic tones in corporate disclosures to accommodate machine processing, adapting to the growing prevalence of AI readership. Firms commonly seek to eschew negative language, particularly after the release of specialized dictionaries such as Loughran and McDonald (2011). This feedback effect of technology on corporate disclosure resembles Goodhart’s law and presents a challenge akin to the Lucas Critique for textual data.

²⁷Recently, Battaglia et al. (2024) suggested considering the potential bias in the traditional two-step approach of utilizing textual measures without addressing measurement error. Their proposed one-step strategy for unstructured data analysis could provide a more reliable alternative, addressing this limitation also in macro forecasting research.

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A Tables/Figures

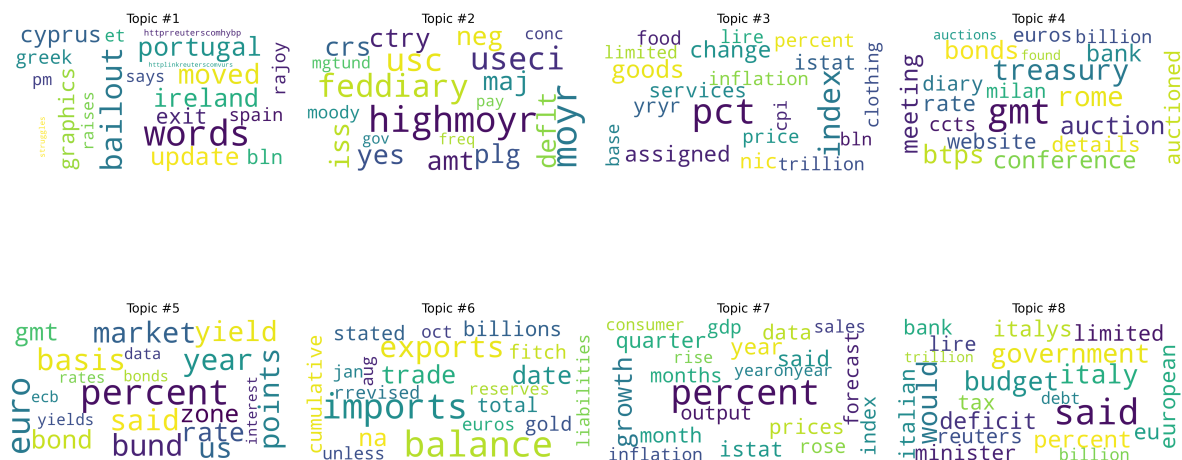
Table A.1: Out-of-sample results: forecasting the monthly survey-based Italian inflation expectations using Twitter-based indicators

	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$
$AR(p)$ ($RMSE$)	6.639	8.971	10.513	11.937	13.283	14.297
$IS1Y_t$	1.095	1.064	1.040	1.004	1.012	1.003
GT_t	1.030	1.003	0.998	0.990	0.994	0.998
TW1-MA(30)	1.042	1.033	1.025	1.024	1.019	1.025
TW2-MA(30)	1.047	1.021	1.009	0.992	0.980	0.961
TW3-(Exp-opt)	1.048	1.031	1.028	1.021	1.017	1.023
TW4-MA(30)	1.050	1.034	1.049	1.023	1.021	1.026
TW1-MA(60)	1.043	1.032	1.028	1.020	1.018	1.025
TW2-MA(60)	1.065+	1.023	1.022	1.003	1.000	0.991
TW3-(Exp-0.3)	1.043	1.025	1.020	1.019	1.017	1.021
TW4-MA(60)	1.046	1.027	1.023	1.021	1.022	1.024
TW1-MA(30)-Eco	1.056	1.159	1.123	1.073	1.063	1.035
TW2-MA(30)-Eco	0.987	0.979	0.960	0.936	0.911	0.881
TW3-(Exp-opt)-Eco	1.060	1.161	1.125	1.088	1.068	1.051
TW4-MA(30)-Eco	1.061	1.180	1.141	1.103	1.070	1.060
TW1-MA(60)-Eco	1.052	1.135	1.119	1.071	1.067	1.035
TW2-MA(60)-Eco	0.998	0.974	0.959	0.942	0.901	0.892
TW3-(Exp-0.3)-Eco	1.053	1.104	1.085	1.066	1.037	1.097
TW4-MA(60)-Eco	1.056	1.129	1.099	1.080	1.045	1.110
TW1-MA(30)-News	1.088	1.042	1.057	1.035	1.042	1.049
TW2-MA(30)-News	1.053	1.037	1.025	1.002	1.003	0.976
TW3-(Exp-opt)-News	1.081	1.043	1.053	1.031	1.042	1.043
TW4-MA(30)-News	1.081	1.045	1.054	1.031	1.044	1.045
TW1-MA(60)-News	1.095	1.043	1.059	1.039	1.041	1.053
TW2-MA(60)-News	1.056	1.021	1.022	1.007	1.002	0.996
TW3-(Ex-0.3)-News	1.071	1.041	1.055	1.026	1.034	1.031
TW4-MA(60)-News	1.073	1.044	1.038	1.026	1.036	1.032

Note: The table presents the RMSE for the benchmark $AR(p)$ model (first row) and the ratio of the RMSE of each competing model with respect to the benchmark. A number below 1 means that the competing model outperforms. Numbers in boldface represent the models with the lowest RMSE for each forecast horizon h (from 1 to 6 months ahead). Recursive scheme with first in-sample of 36 observations. ***, **, and * indicate significance at 1, 5, and 10% respectively of the Diebold-Mariano test of equal forecast accuracy. +++, ++, and + indicate significance at 1, 5, and 10% respectively, when the benchmark outperforms.

Figure A.1: Topics extracted from Reuters economic articles/paragraphs on Italy.

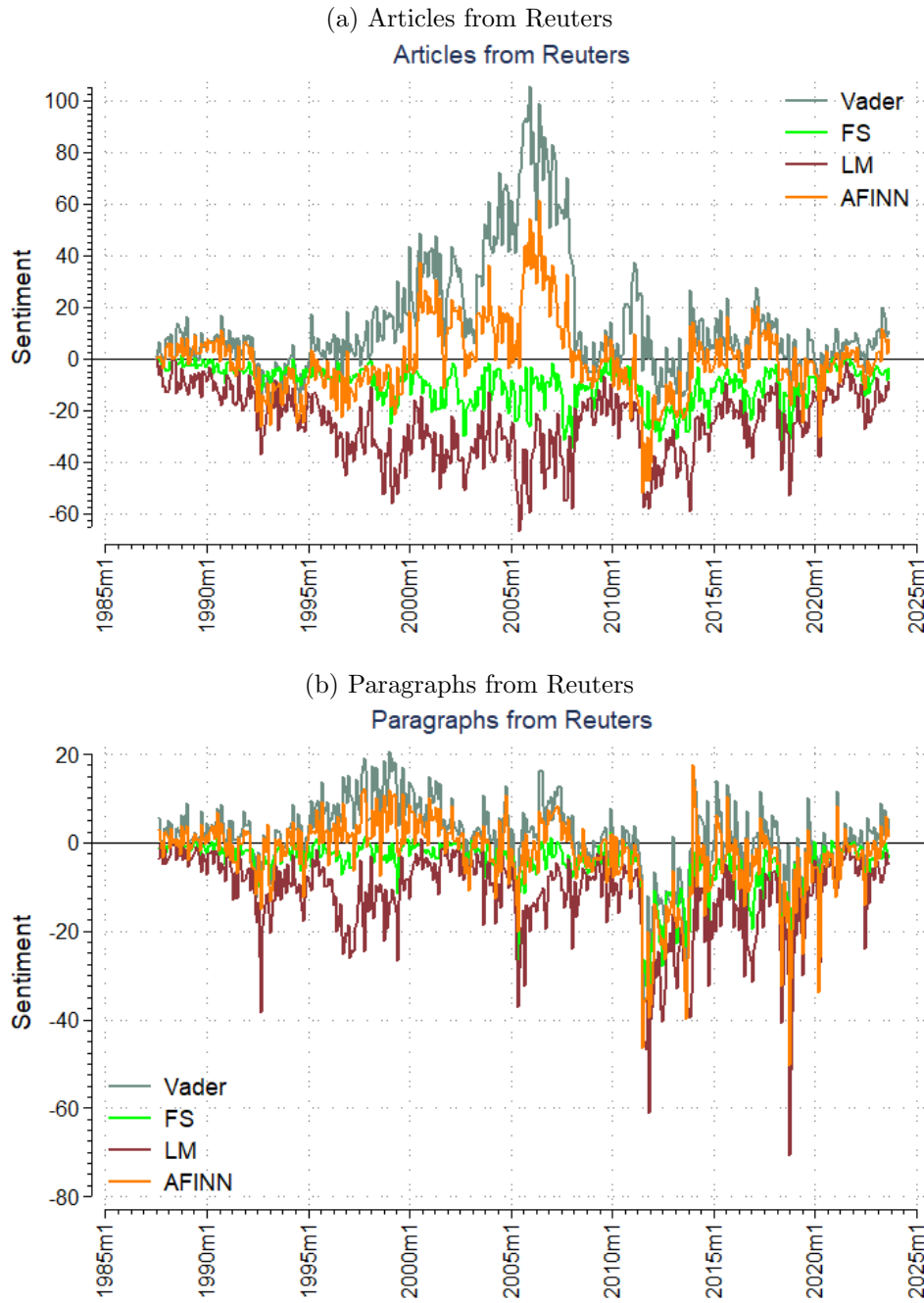
(a) Article-level - Case 1



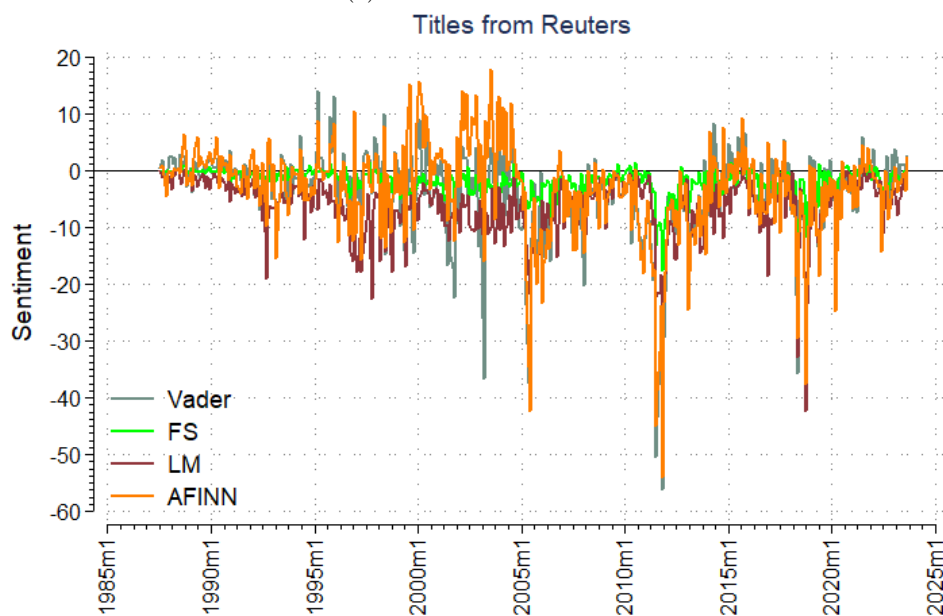
(b) Article-level - Case 2



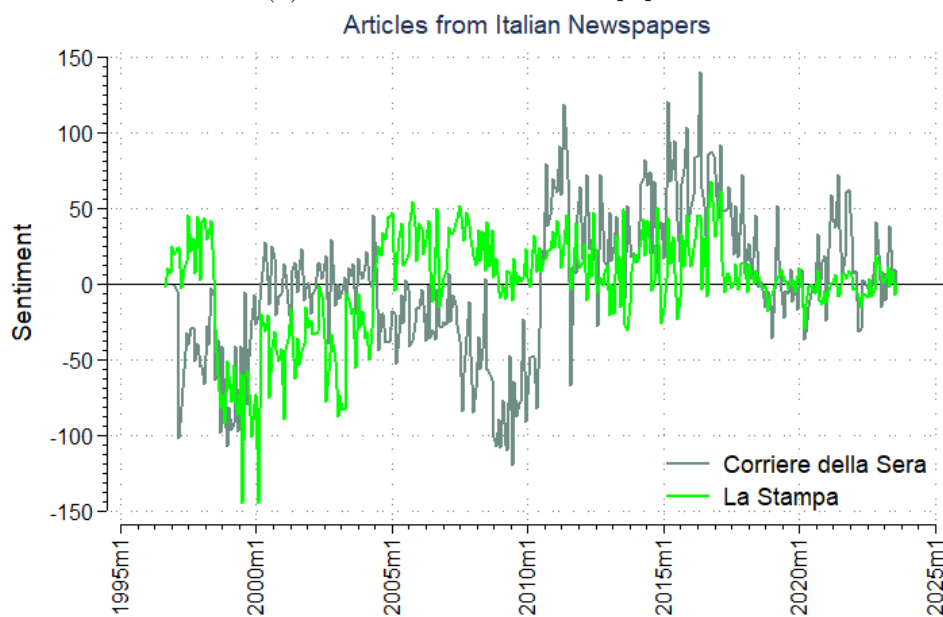
Figure A.2: Sentiment indicators for Italy across dimensions and languages.



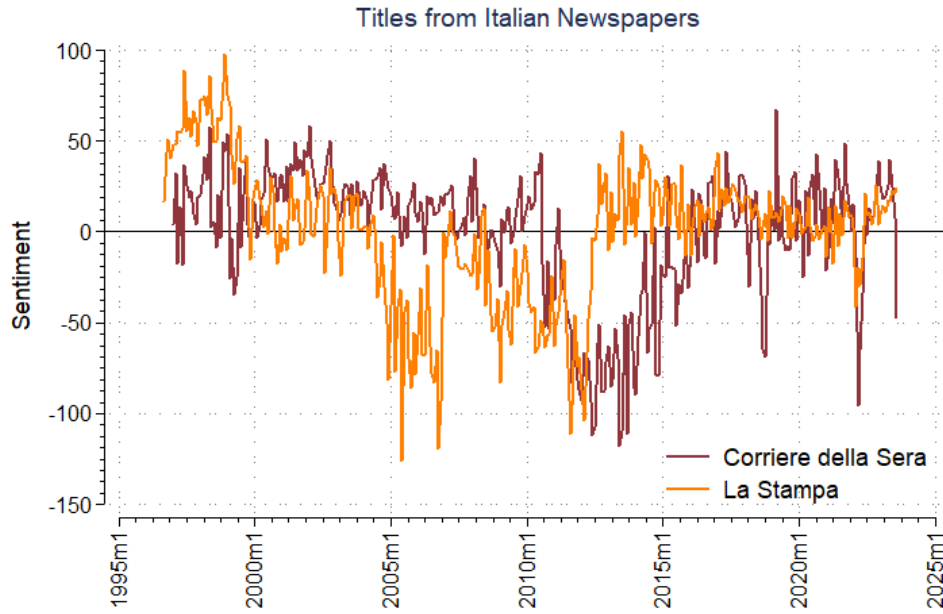
(c) Titles from Reuters



(d) Articles from Italian Newspapers



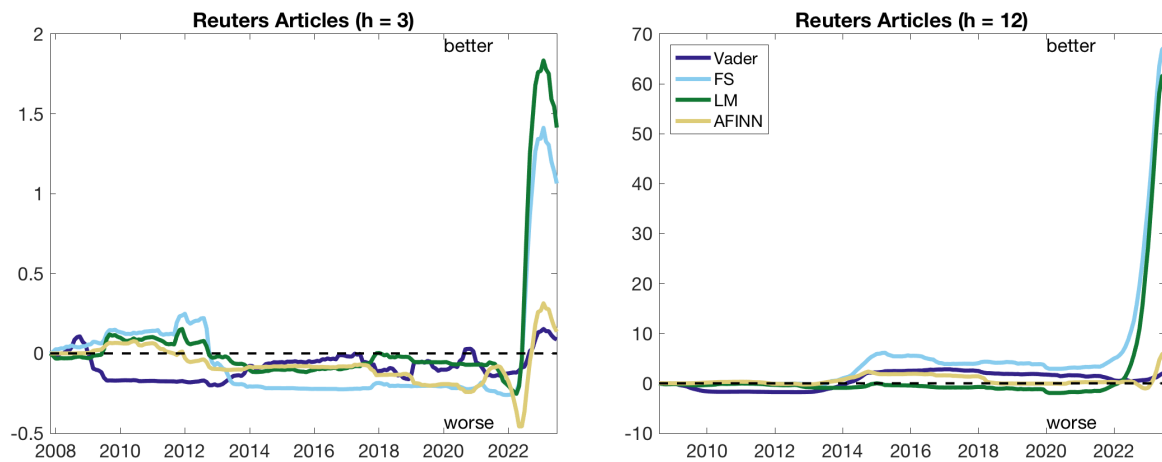
(e) Titles from Italian Newspapers



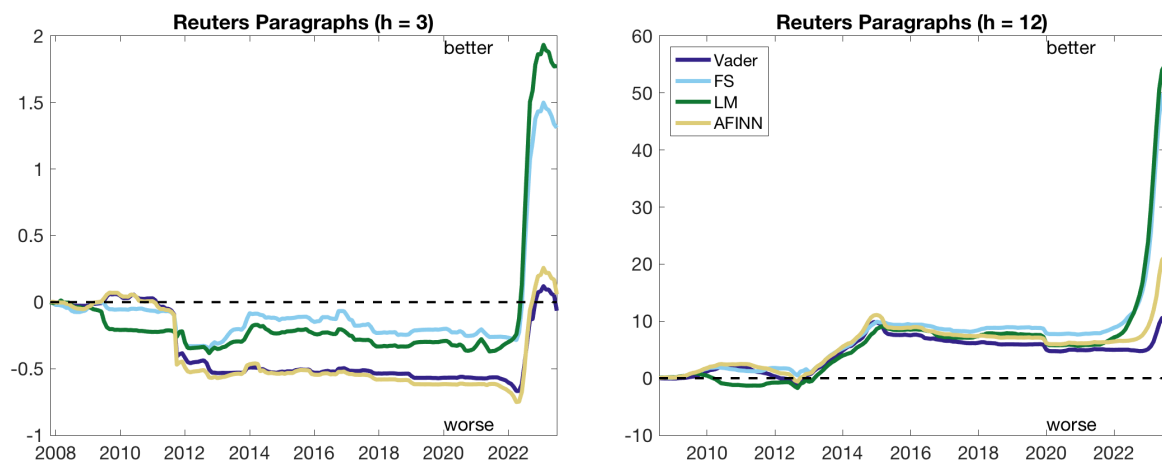
Note: The figure presents monthly time series of 4 sentiment indicators from Reuters articles on Italy at the article ((A.2a)), paragraph ((A.2b)), and title level ((A.2c)). It also includes sentiment from two Italian newspapers (Corriere della Sera and La Stampa) at the article ((A.2d)) and title level ((A.2e)).

Figure A.3: Cumulative Sum of Squared Forecast Errors for inflation

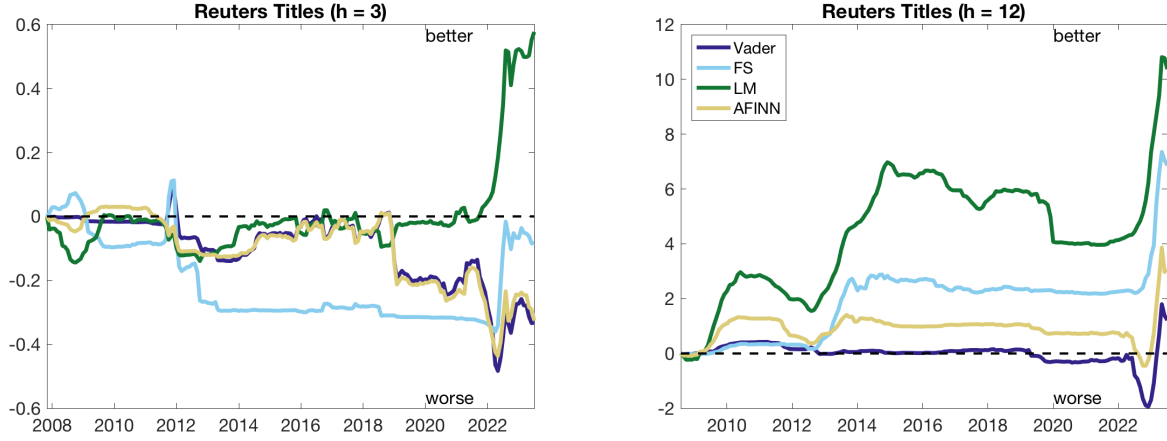
(a) Reuters Articles



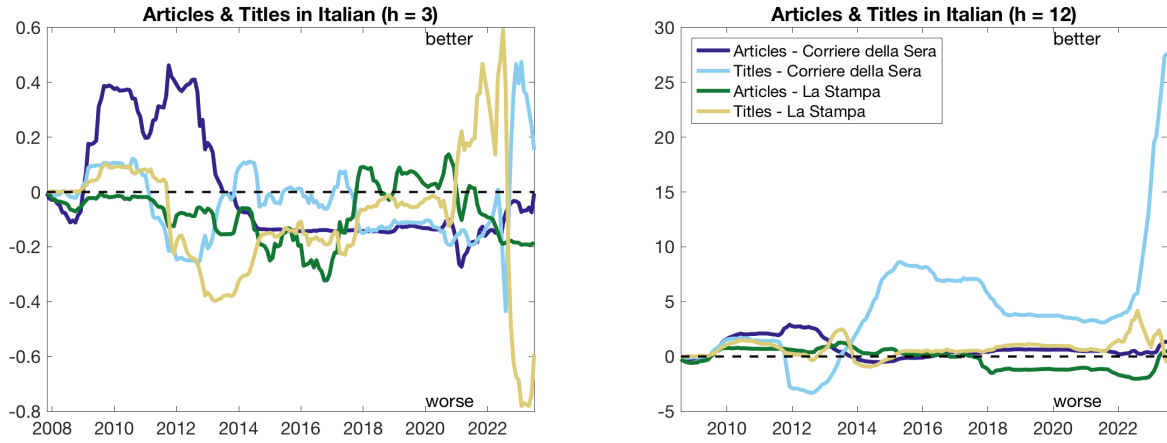
(b) Reuters Paragraphs



(c) Reuters Titles



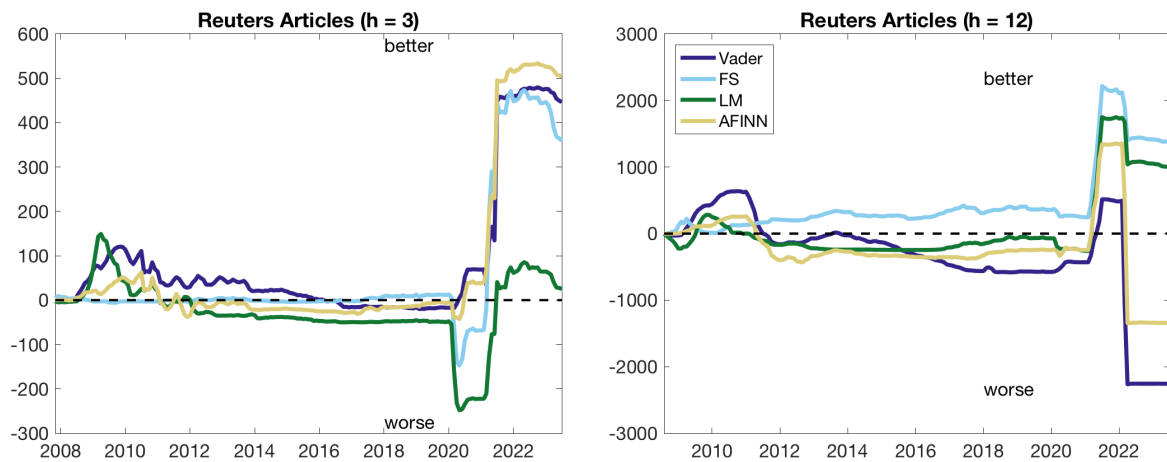
(d) Articles and Titles in Italian



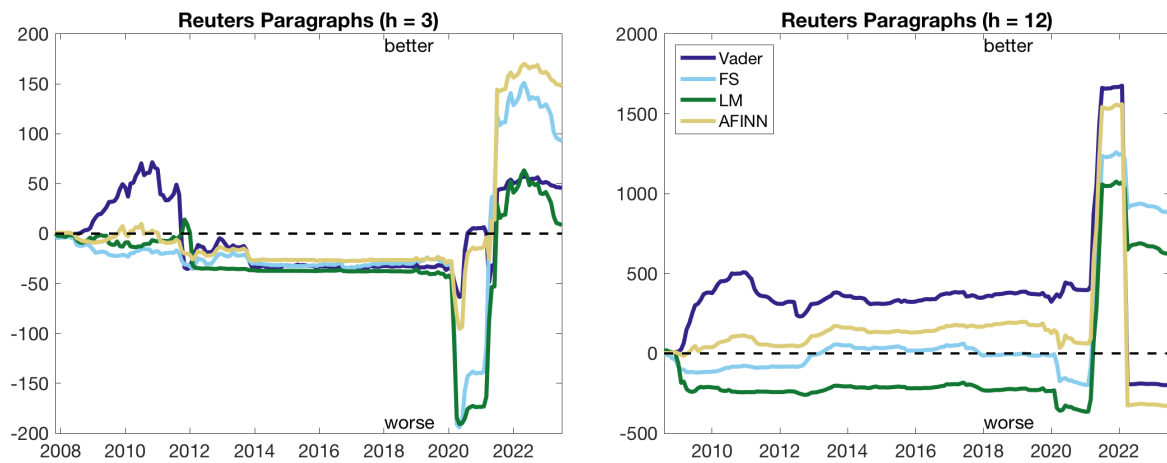
Note: The figure shows CSSED for 3-, and 12-month-ahead inflation forecasts in Italy. Sentiment sources include Reuters articles (A.3a), paragraphs (A.3b), and titles (A.3c) in English, as well as articles and titles from Italian newspapers Corriere della Sera and La Stampa (A.3d). Benchmark: AR(p); Approach: Recursive; In-sample: Jan. 1997 - Jun. 2007; Out-of-sample: Jul. 2007 - Apr. 2023.

Figure A.4: Cumulative Sum of Squared Forecast Errors for industrial production.

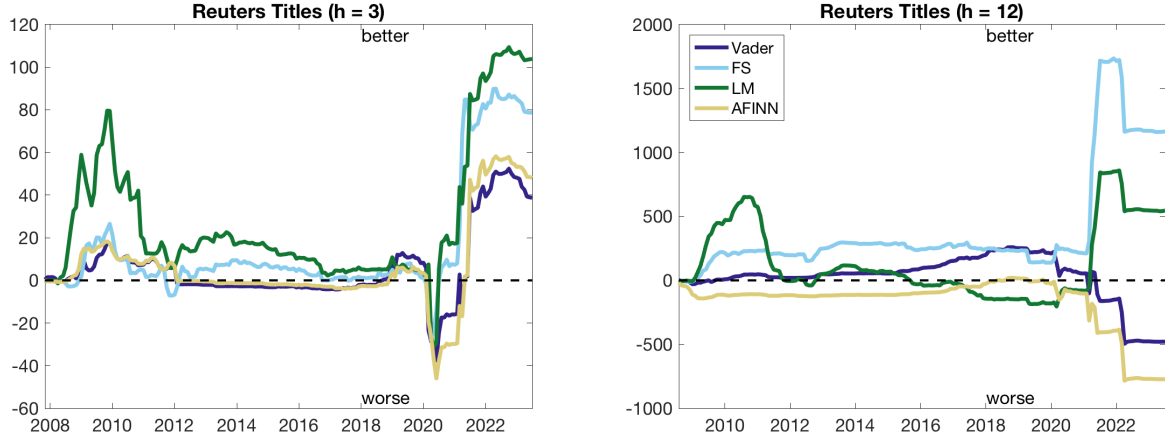
(a) Reuters Articles



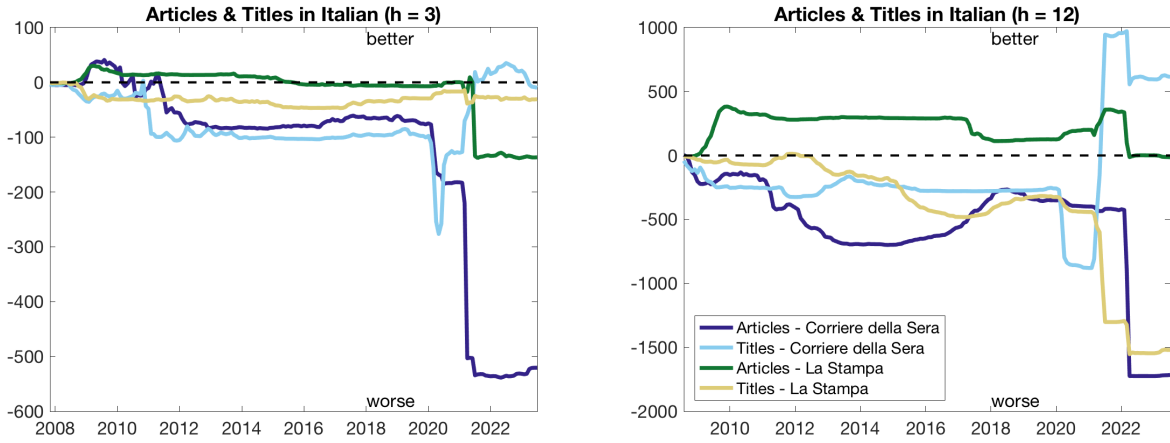
(b) Reuters Paragraphs



(c) Reuters Titles



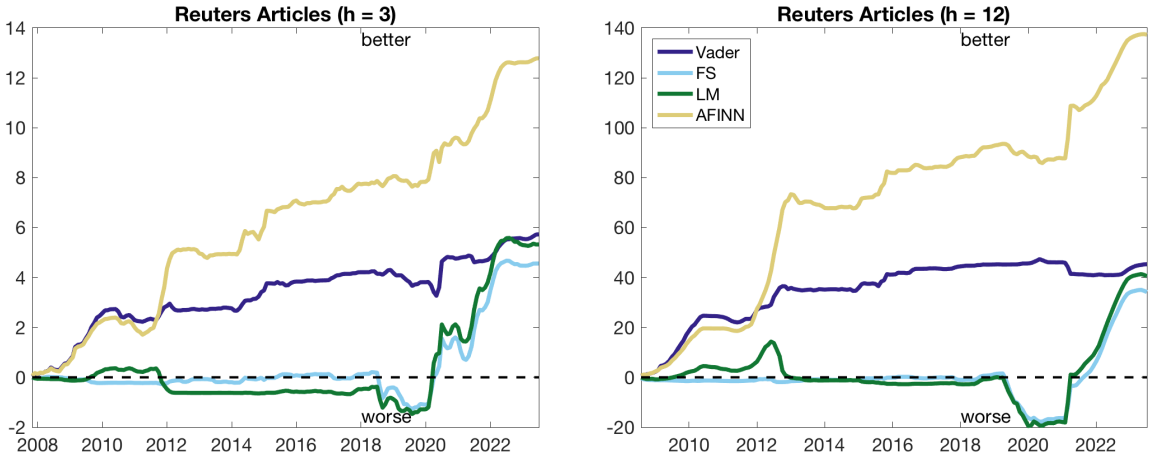
(d) Articles and Titles in Italian



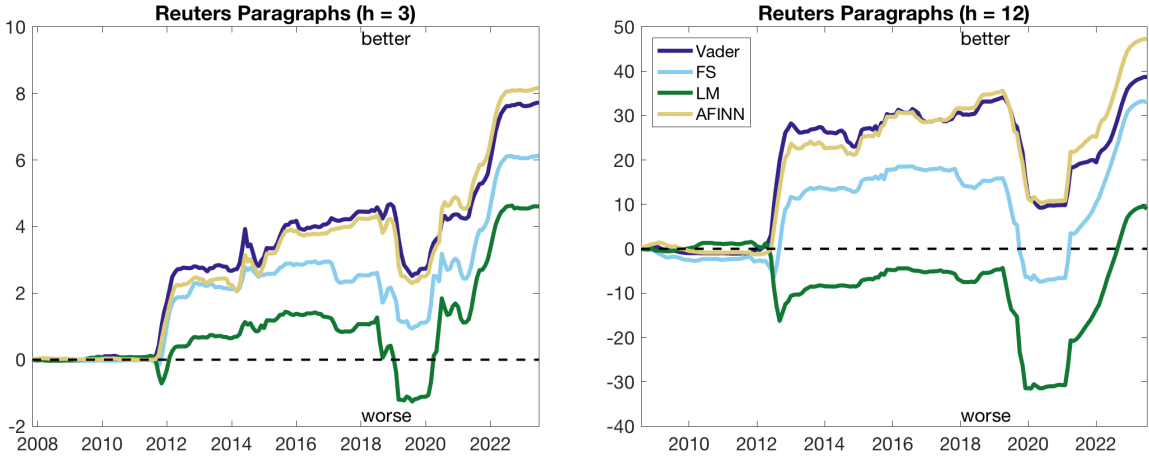
Note: The figure displays the CSSED for the Italian industrial production for 3-, and 12-month-ahead forecasts using sentiment from different sources: Reuters articles (A.4a), paragraphs (A.4b), and titles (A.4c) in English, as well as articles and titles from Italian newspapers *Corriere della Sera* and *La Stampa* (A.4d). Benchmark: AR(p); Scheme: Recursive; In-sample: Jan. 1997 - Jun. 2007; Out-of-sample: Jul. 2007 - Apr. 2023.

Figure A.5: Cumulative Sum of Squared Forecast Errors for unemployment rate.

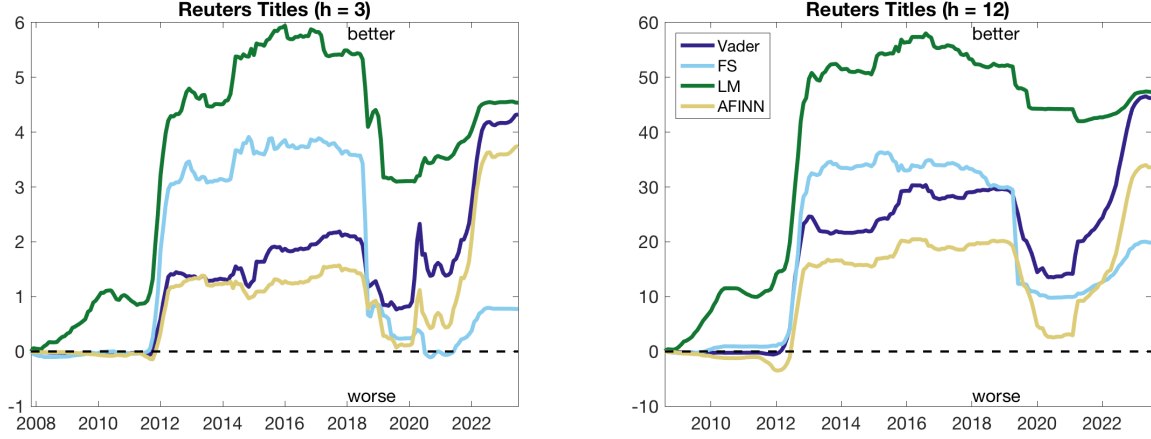
(a) Reuters Articles



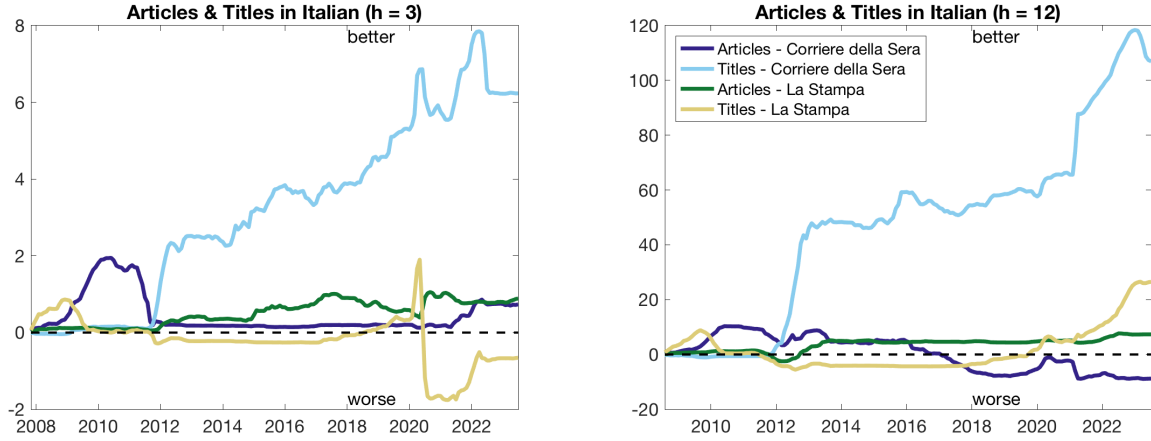
(b) Reuters Paragraphs



(c) Reuters Titles

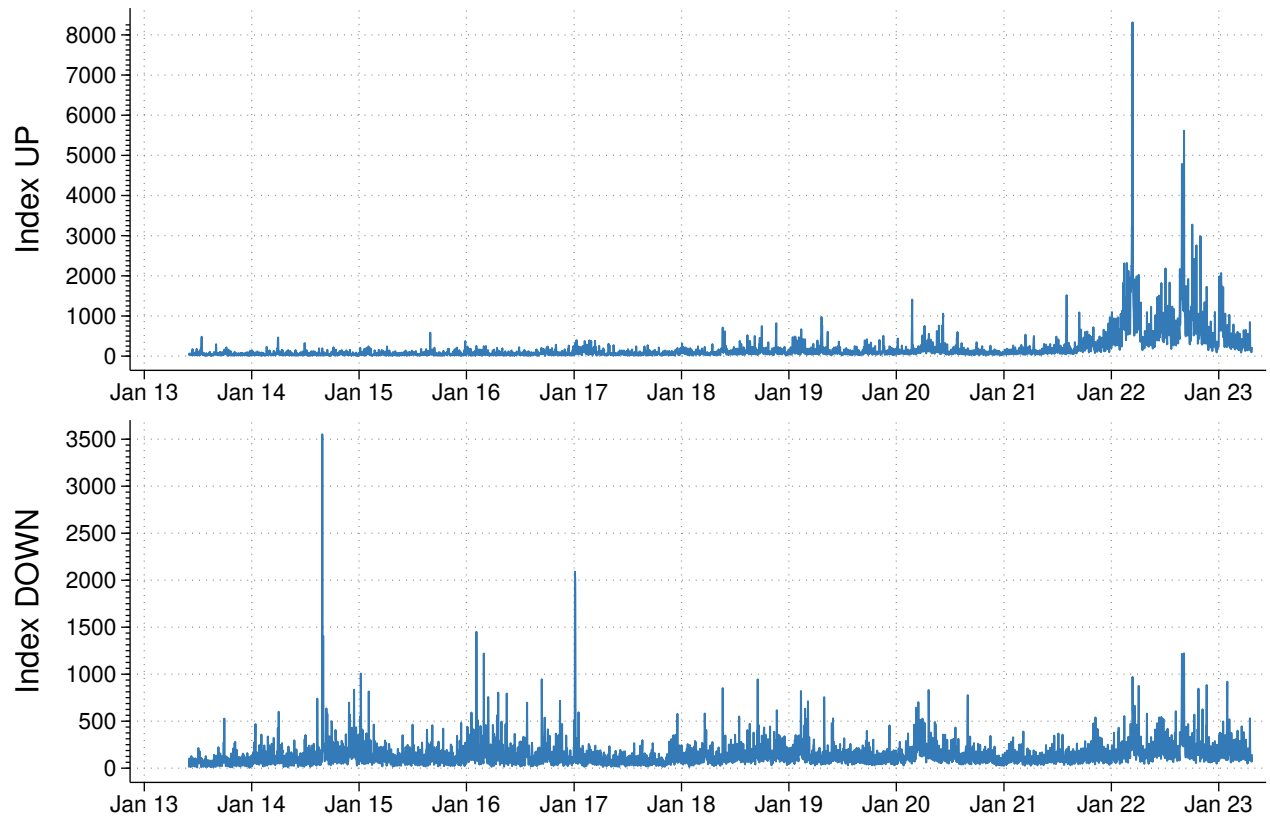


(d) Articles and Titles in Italian



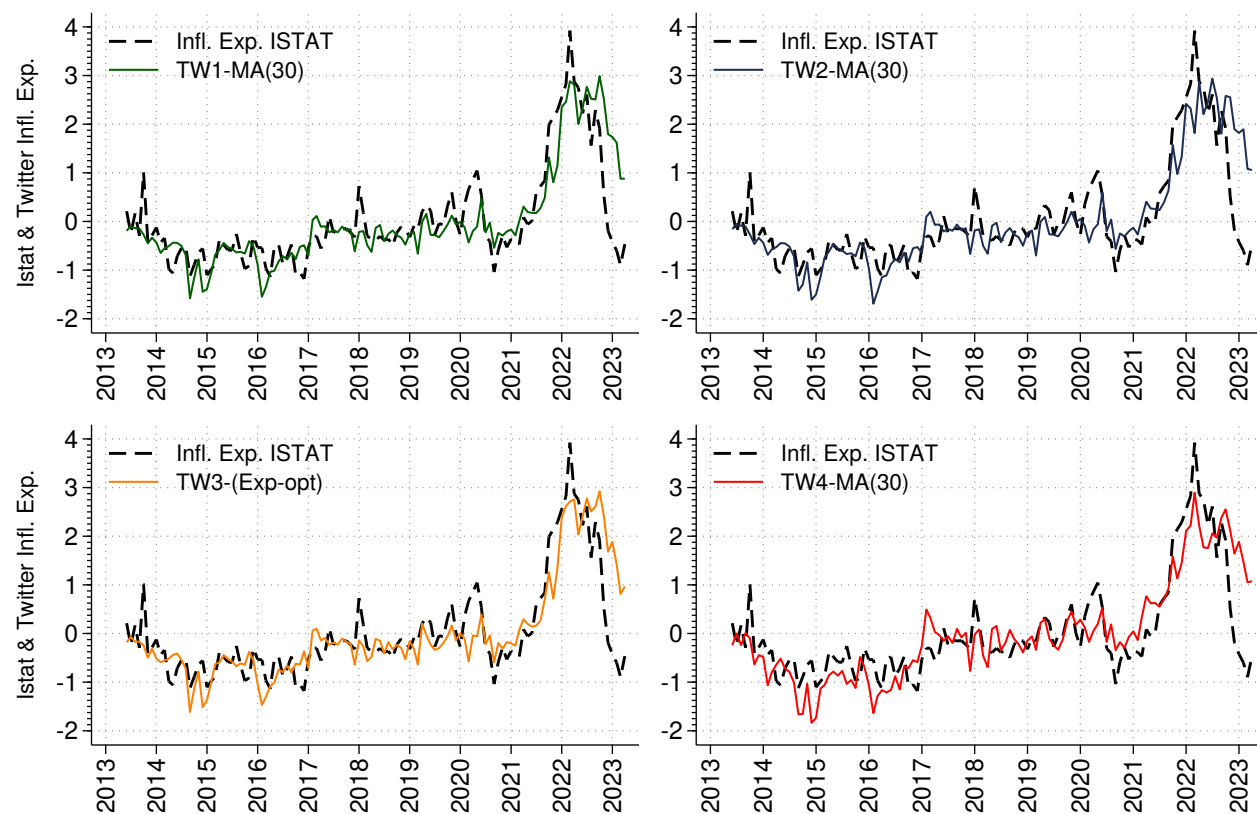
Note: The figure displays the CSSED for the Italian unemployment rate for 3-, and 12-month-ahead forecasts using sentiment from different sources: Reuters articles (A.5a), paragraphs (A.5b), and titles (A.5c) in English, as well as articles and titles from Italian newspapers *Corriere della Sera* and *La Stampa* (A.5d). Benchmark: AR(p); Scheme: Recursive; In-sample: Jan. 1997 - Jun. 2007; Out-of-sample: Jul. 2007 - Apr. 2023.

Figure A.6: Dictionary-based Inflation Expectation Indexes - Up and Down indexes



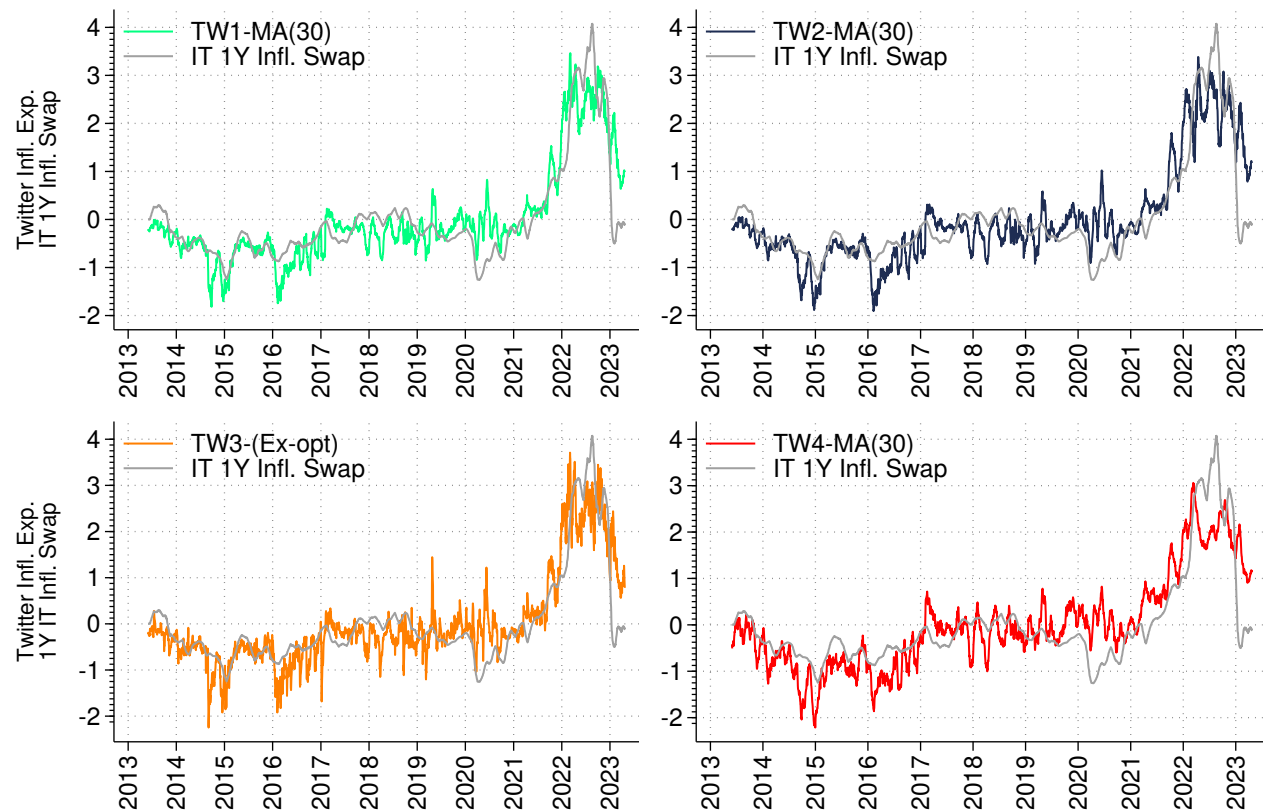
Note: The figure depicts the two dictionary-based Up and Down indexes based on the daily volume of tweets.

Figure A.7: Comparison of Twitter-based inflation expectation indicators and surveys for Italy.



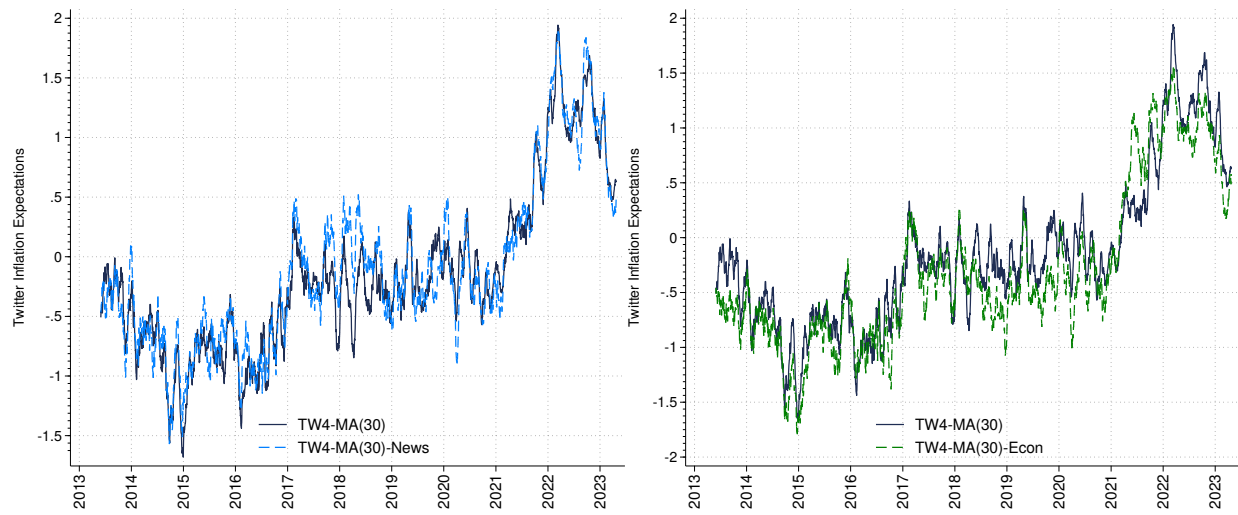
Note: The figure displays the four Twitter-based inflation expectation indexes for Italy compared with the survey-based expectations from ISTAT. All indicators are standardized.

Figure A.8: Twitter-based Inflation Expectations Indicators vs Italian 1Y inflation swap



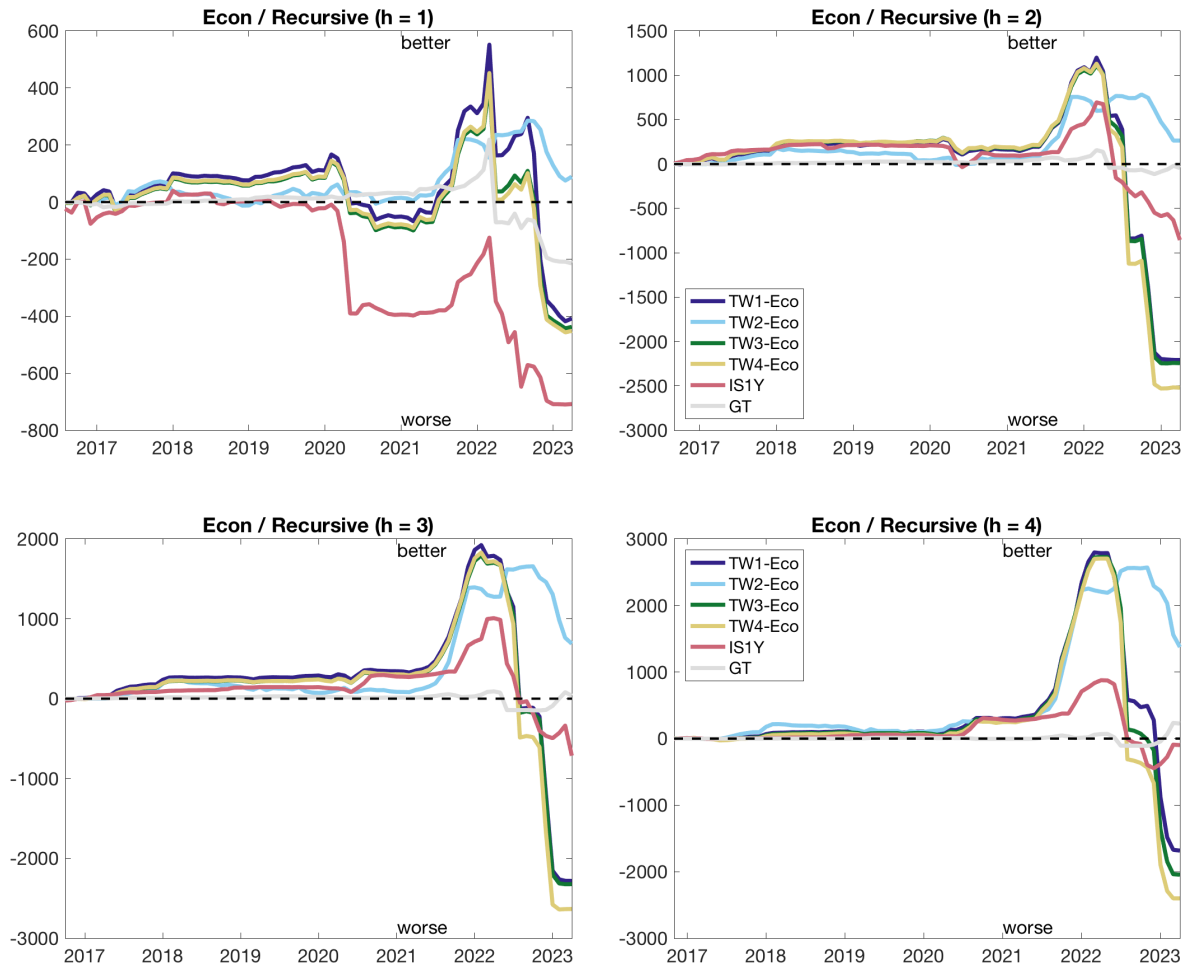
Note: The figure displays the four Twitter-based inflation expectation indicators vs the 1Y inflation swap contract linked to the Italian inflation (IT Infl. Swap 1Y). All indicators are standardized.

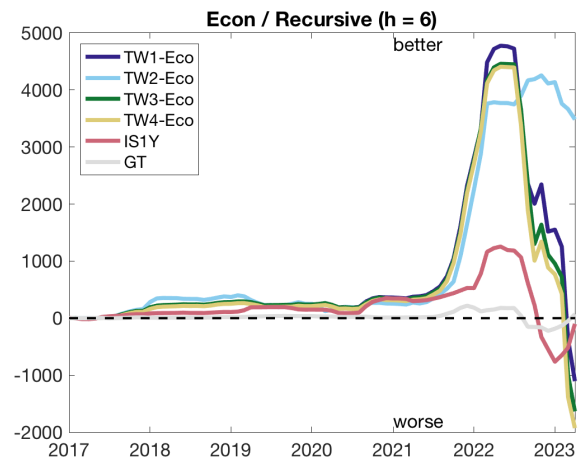
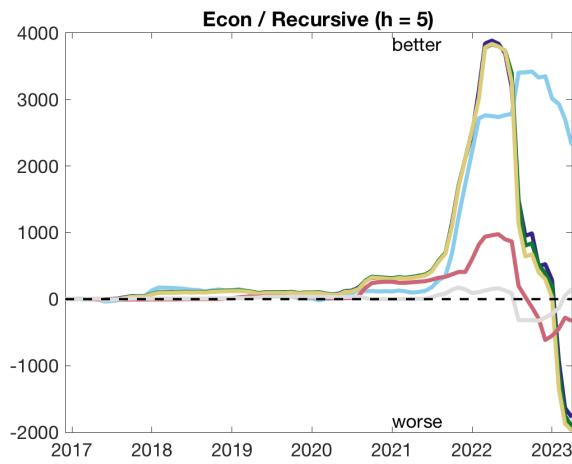
Figure A.9: Twitter-based Inflation Expectations Indicators with sub-samples of users interested in news and economics



Note: Data are at daily frequency, from 1 June 2013 through 30 April 2023. Twitter-based inflation expectation indicators are computed on the “News” (left-panel) or the “Econ” (right-panel) sub-samples.

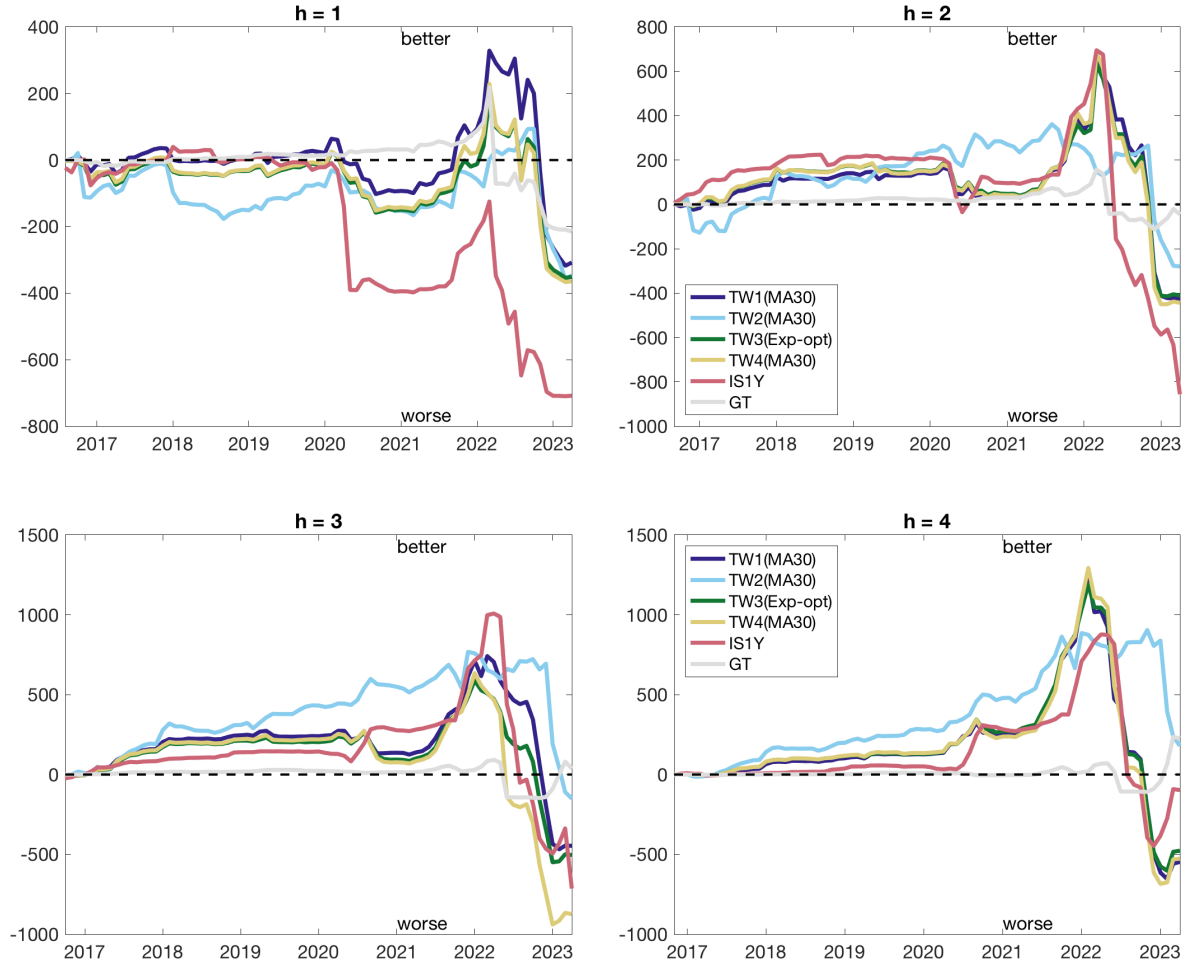
Figure A.10: Out-of-sample comparison: Cumulative Sum of Squared Error Differences with ‘Econ’
Twitter-based indicators

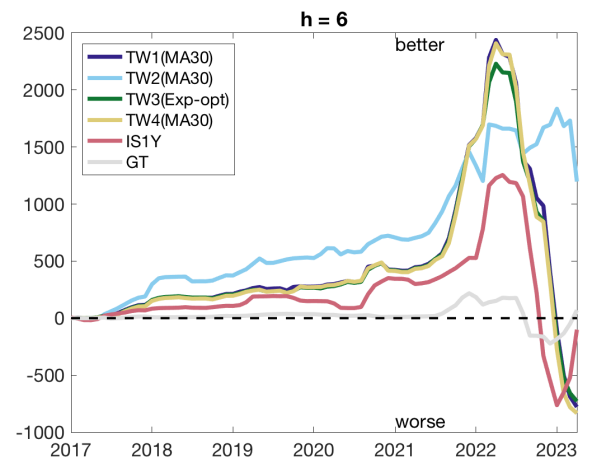
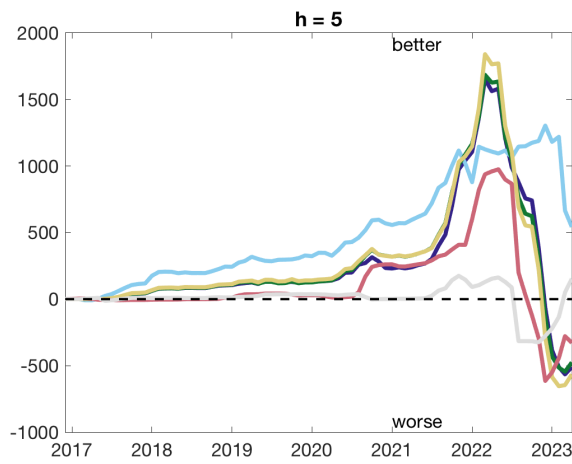




Note: $CSSED_{m,\tau} = \sum_{\tau=R}^T (\hat{e}_{bm,\tau}^2 - \hat{e}_{m,\tau}^2)$. CSSED is below 1 if the $AR(p)$ benchmark outperforms the competing model and above 1 if the competing model fares better. The figures report the CSSED for the four Twitter-based indicators with a backward-looking MA of 30 days computed among users interested in economics, the one-year inflation swap (*IS1Y*), and the Google Trends (*GT*) indicator. Scheme: Recursive; In-sample: Jun. 2013 - May 2016; Out-of-sample: Jun. 2016 - Apr. 2023.

Figure A.11: Out-of-sample comparison: Cumulative Sum of Squared Error Differences with Twitter-based indicators





Note: $CSSED_{m,\tau} = \sum_{\tau=R}^T (\hat{e}_{bm,\tau}^2 - \hat{e}_{m,\tau}^2)$. CSSED is below 1 if the $AR(p)$ benchmark outperforms the competing model and above 1 if the competing model fares better. The figures report the CSSED for the four Twitter-based indicators with a backward-looking MA of 30 days computed among all users, the one-year inflation swap (*IS1Y*), and the Google Trends (*GT*) indicator. Scheme: Recursive; In-sample: Jun. 2013 - May 2016; Out-of-sample: Jun. 2016 - Apr. 2023.